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Watariuchi

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(54) **INFORMATION PROCESSING APPARATUS,
INFORMATION PROCESSING SYSTEM,
AND METHOD FOR INFORMATION
PROCESSING APPARATUS**

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H04N 2201/0072 (2013.01); **H04N 2201/0094**
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USPC 399/75–85, 8–16; 714/47.1–57;
715/733–751
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Division

(57) **ABSTRACT**

A configuration is provided in which in a case where a user
operation is performed on an image forming apparatus or a
data processing apparatus while the image forming appara-
tus and the data processing apparatus are operating in
cooperation with each other, the automatic clearing timers of
both the apparatuses are reset.

12 Claims, 22 Drawing Sheets

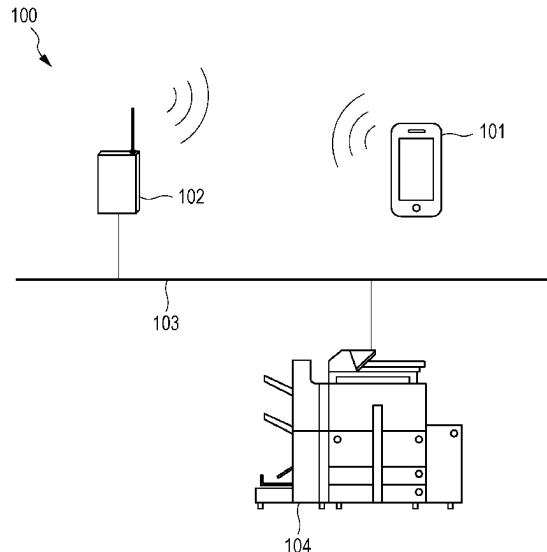


FIG. 1

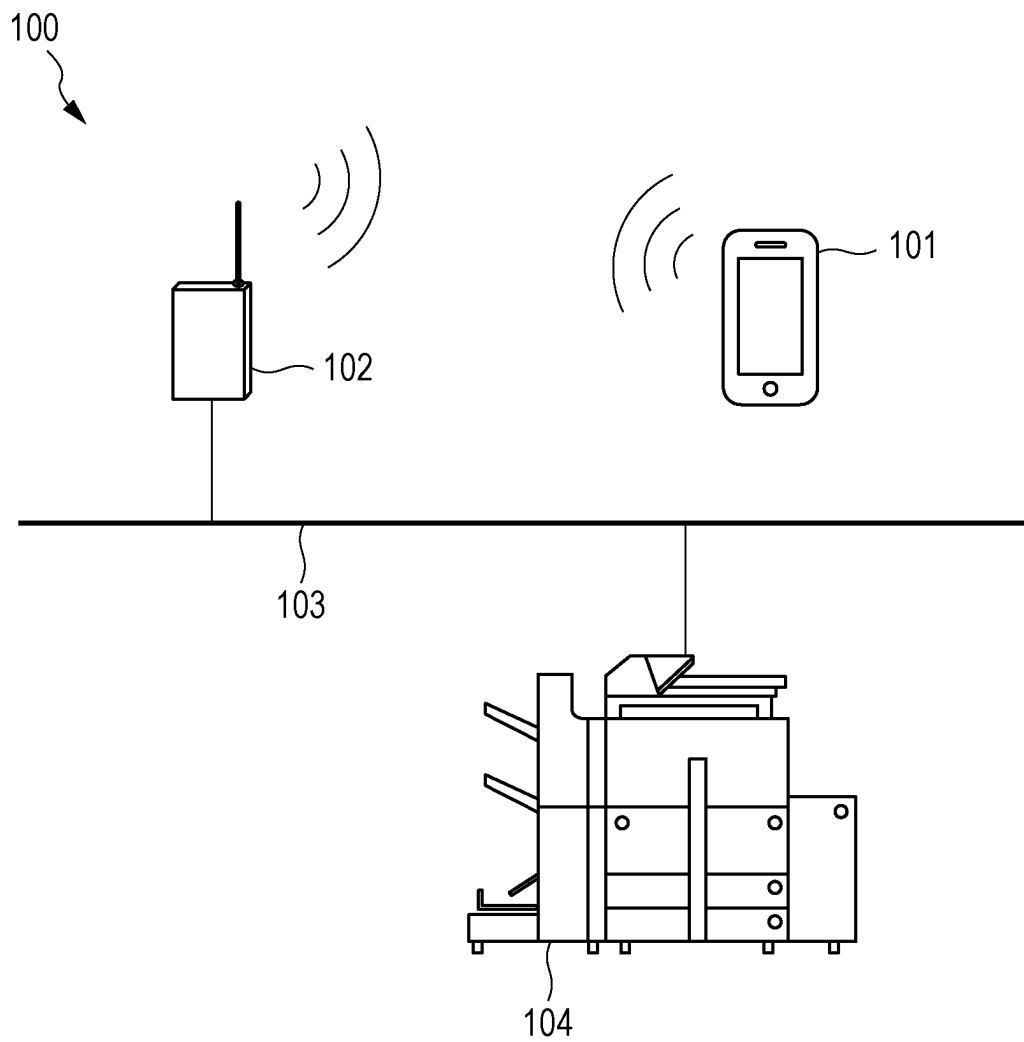


FIG. 2

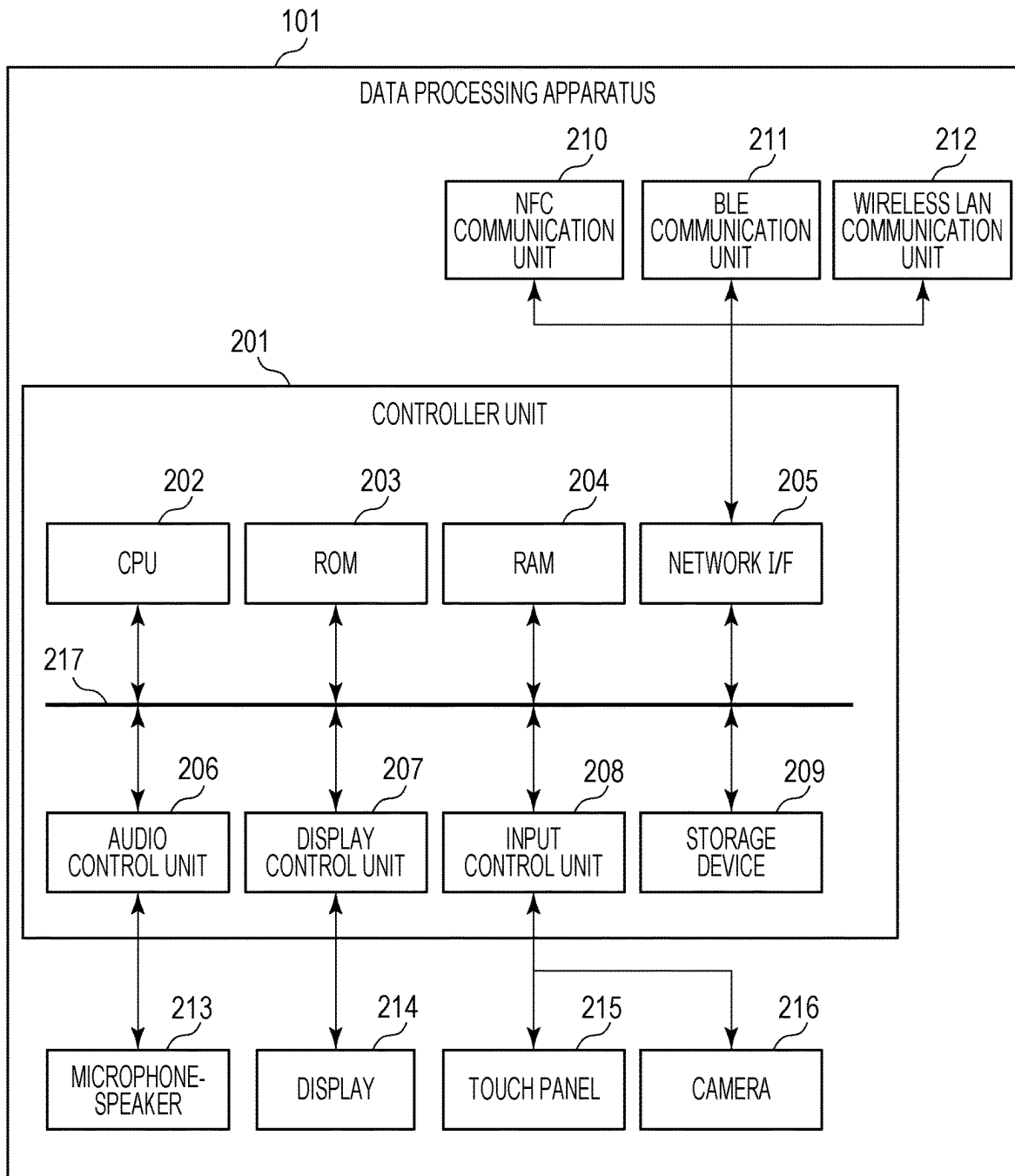


FIG. 3

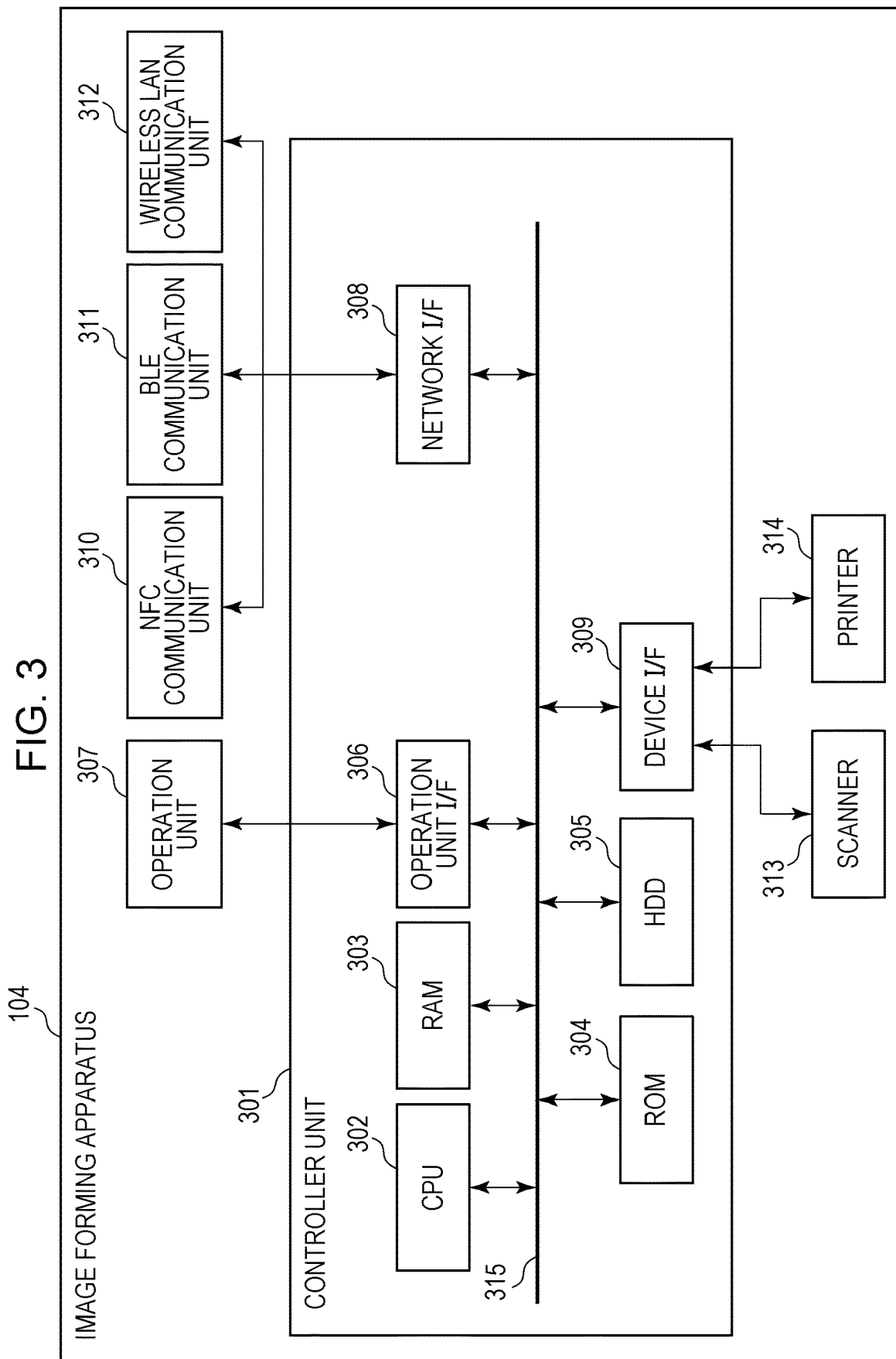


FIG. 4

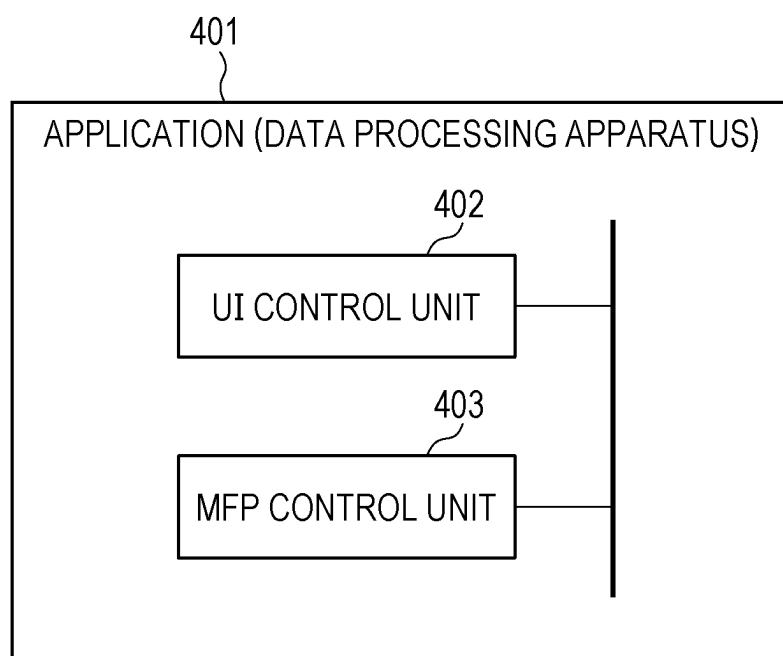


FIG. 5

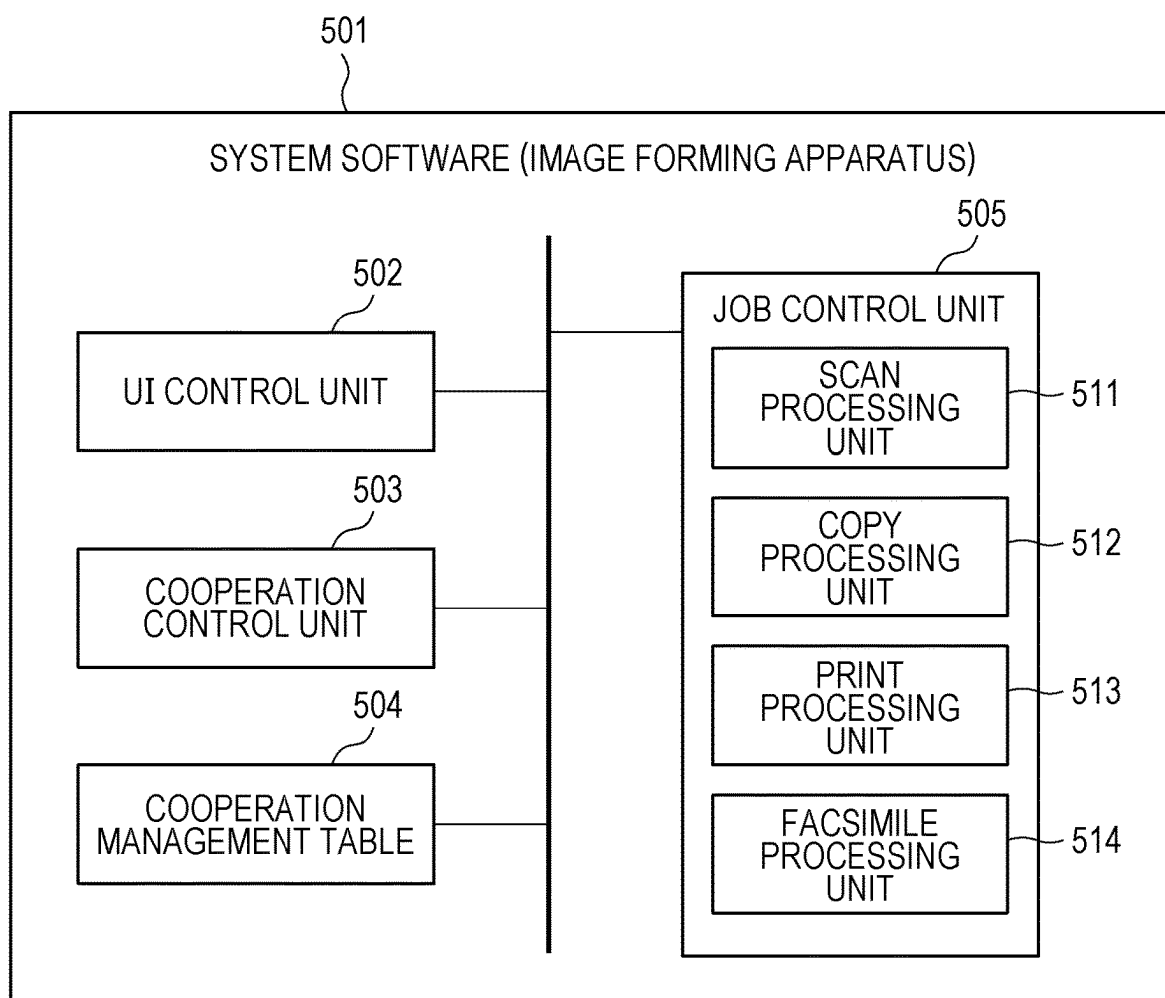


FIG. 6A

600

	601 Request type	602 Cooperation state for acceptance	603 User-operation-causing request
611	Job execution request	(Not in cooperation)	YES
612	Cooperation state obtaining request	(In cooperation)	NO
613	Original-document placing request	"Waiting for an original document to be placed"	YES
614	Unlock-with-PIN request	"PIN code request"	YES
615	Cancel request	Other than "cancel"	YES

FIG. 6B

650

	651 Cooperation state	652 Occurring event	653 Changed cooperation state
661	(Not in cooperation)	—	—
662	Waiting for an original document to be placed	"Scanner: an original document placed", "UI: the placing completed button pressed"	PIN code request
663		"UI: the cancel button pressed"	Cancel
664	PIN code request	"Scanner: no original document placed"	Waiting for an original document to be placed
665		"UI: the cancel button pressed"	Cancel
666	Job execution	—	—
667	Cancel	—	—

FIG. 7

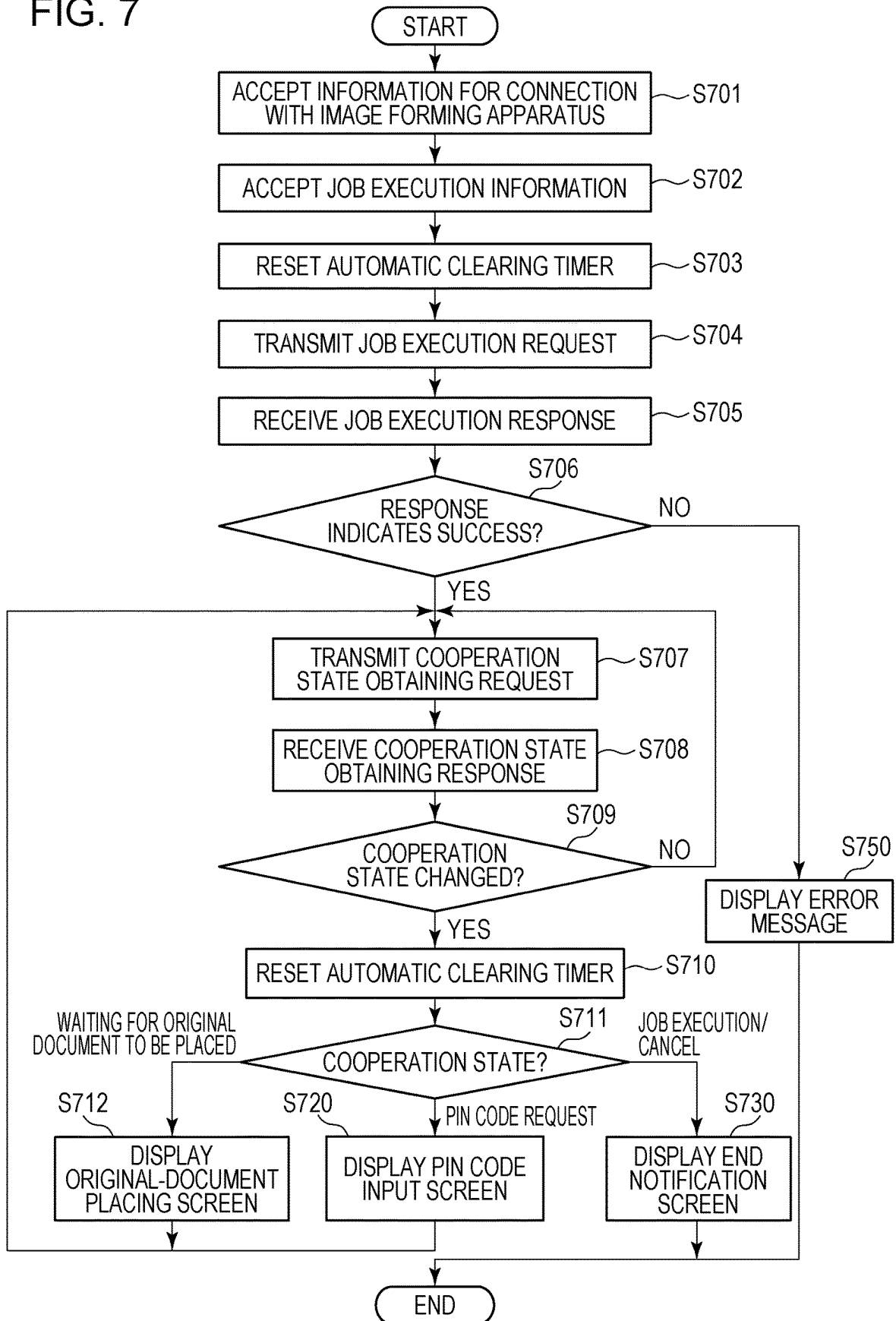


FIG. 8A

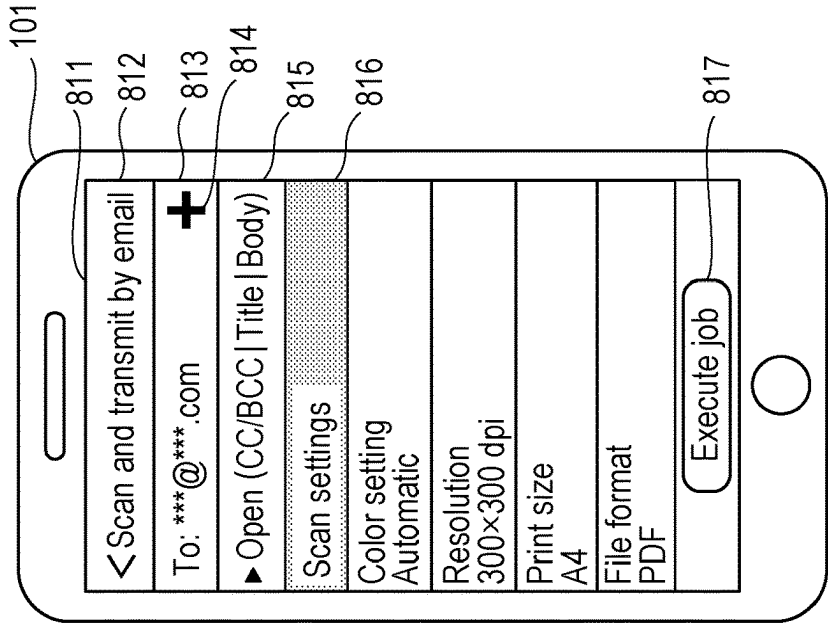


FIG. 8B

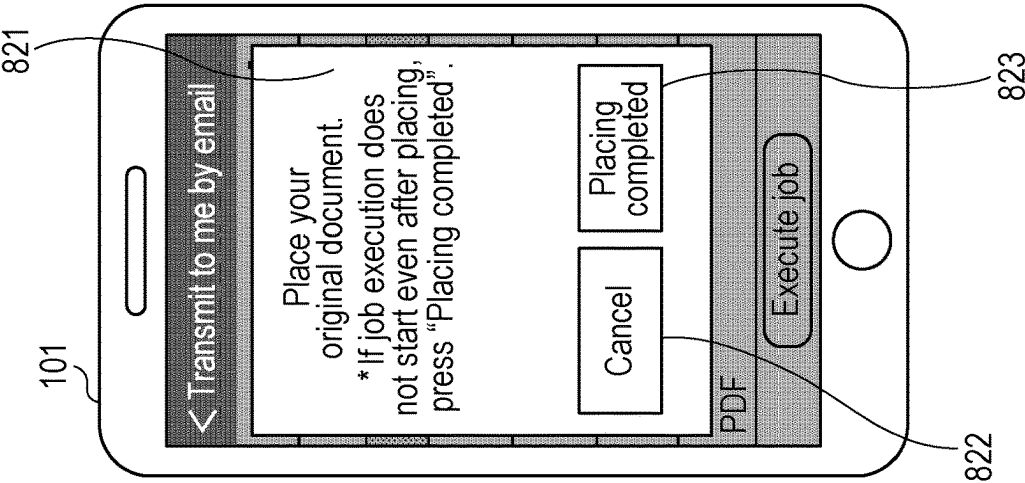


FIG. 8C

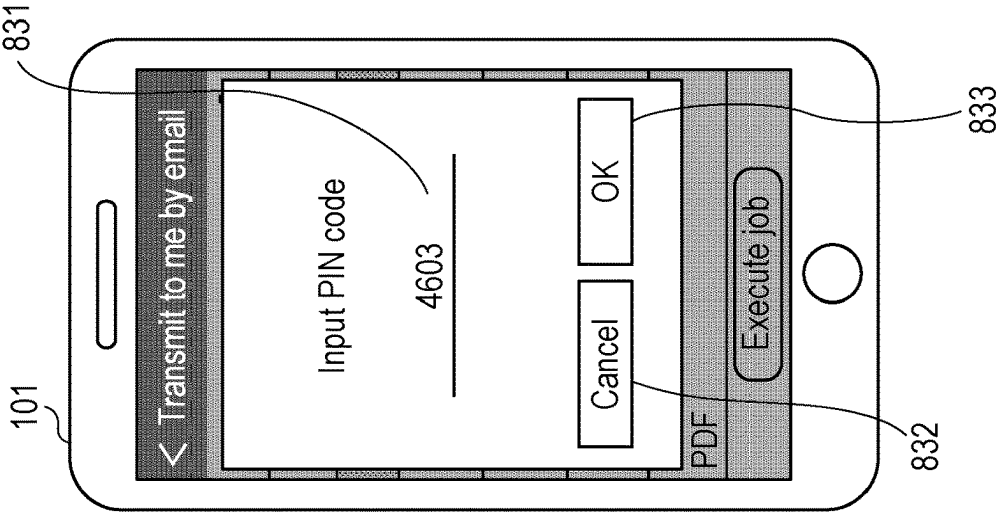


FIG. 8D

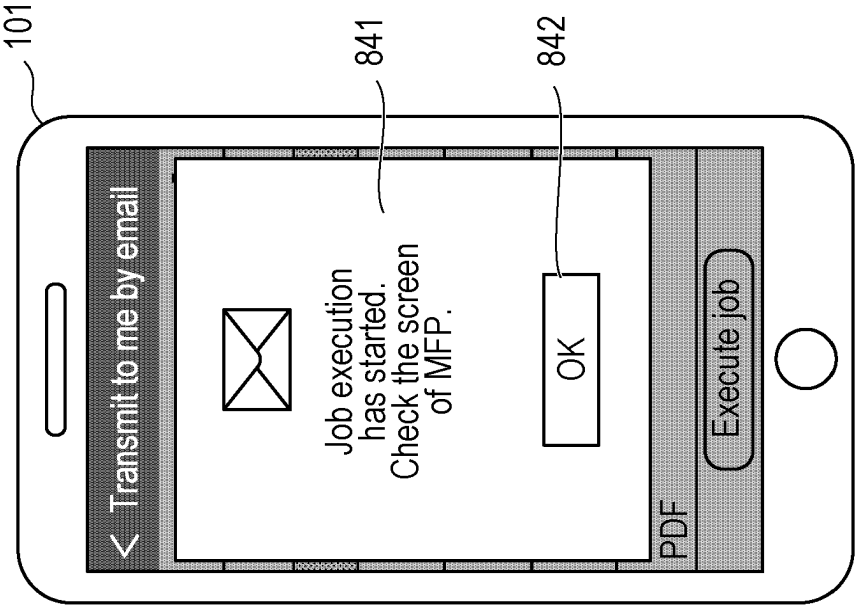


FIG. 8E

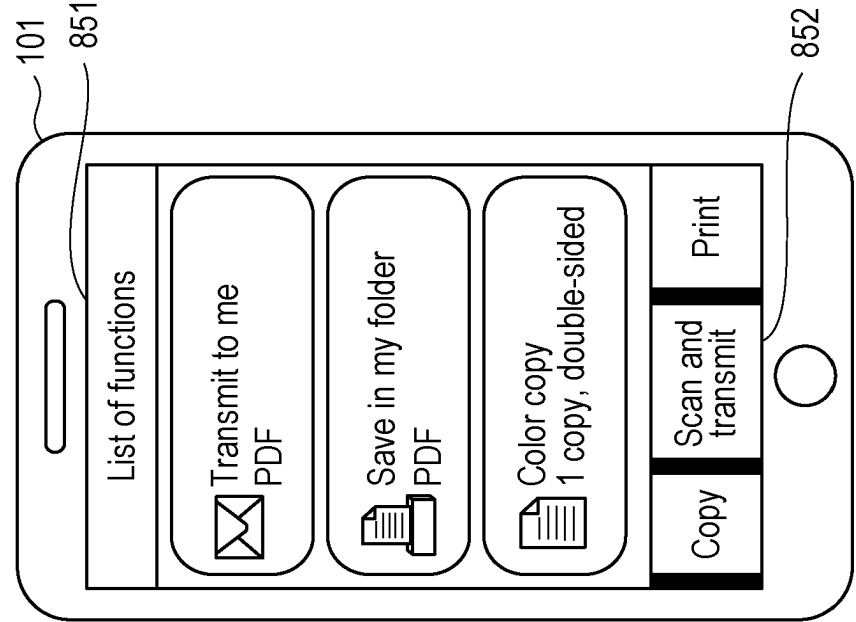


FIG. 9A

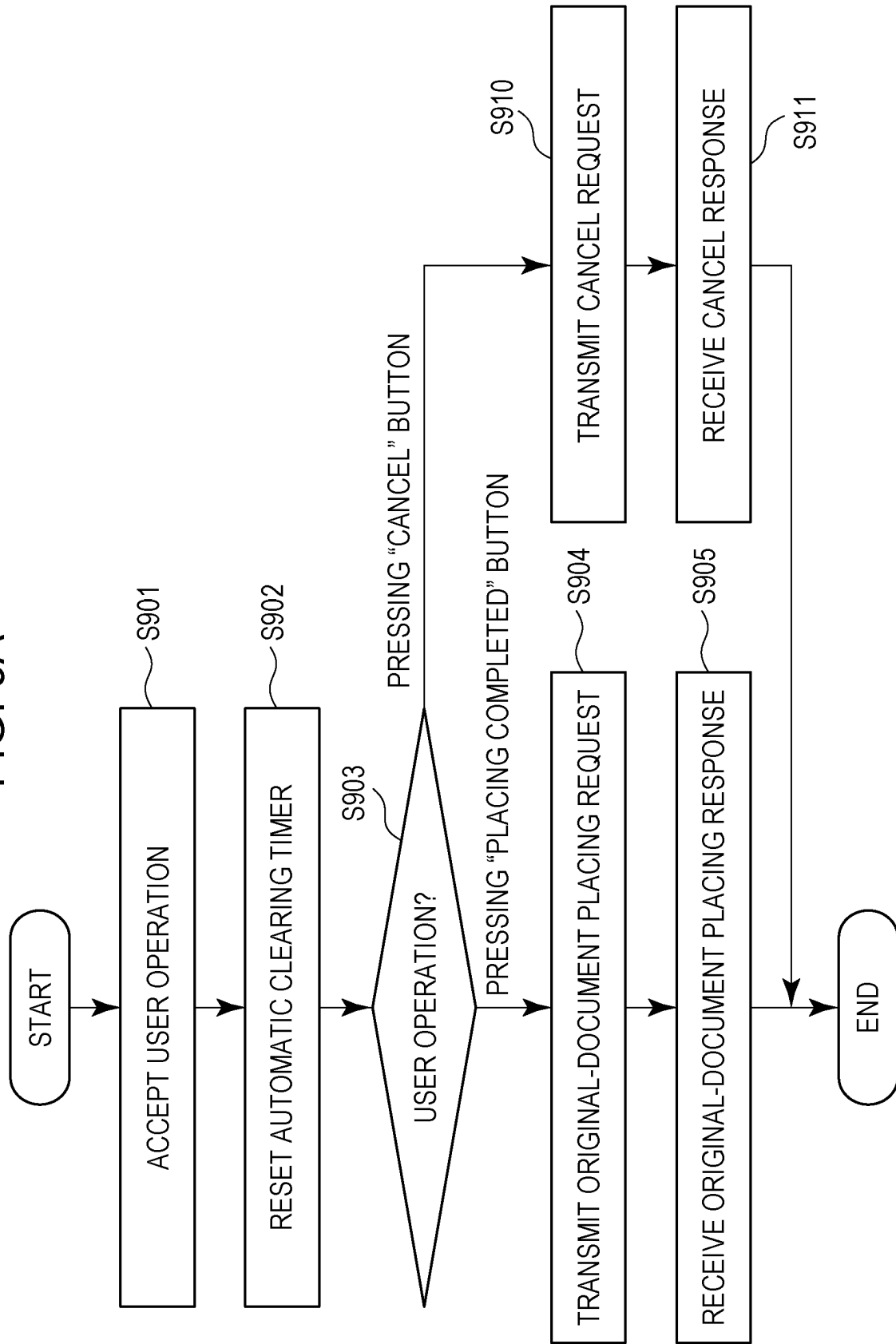


FIG. 9B

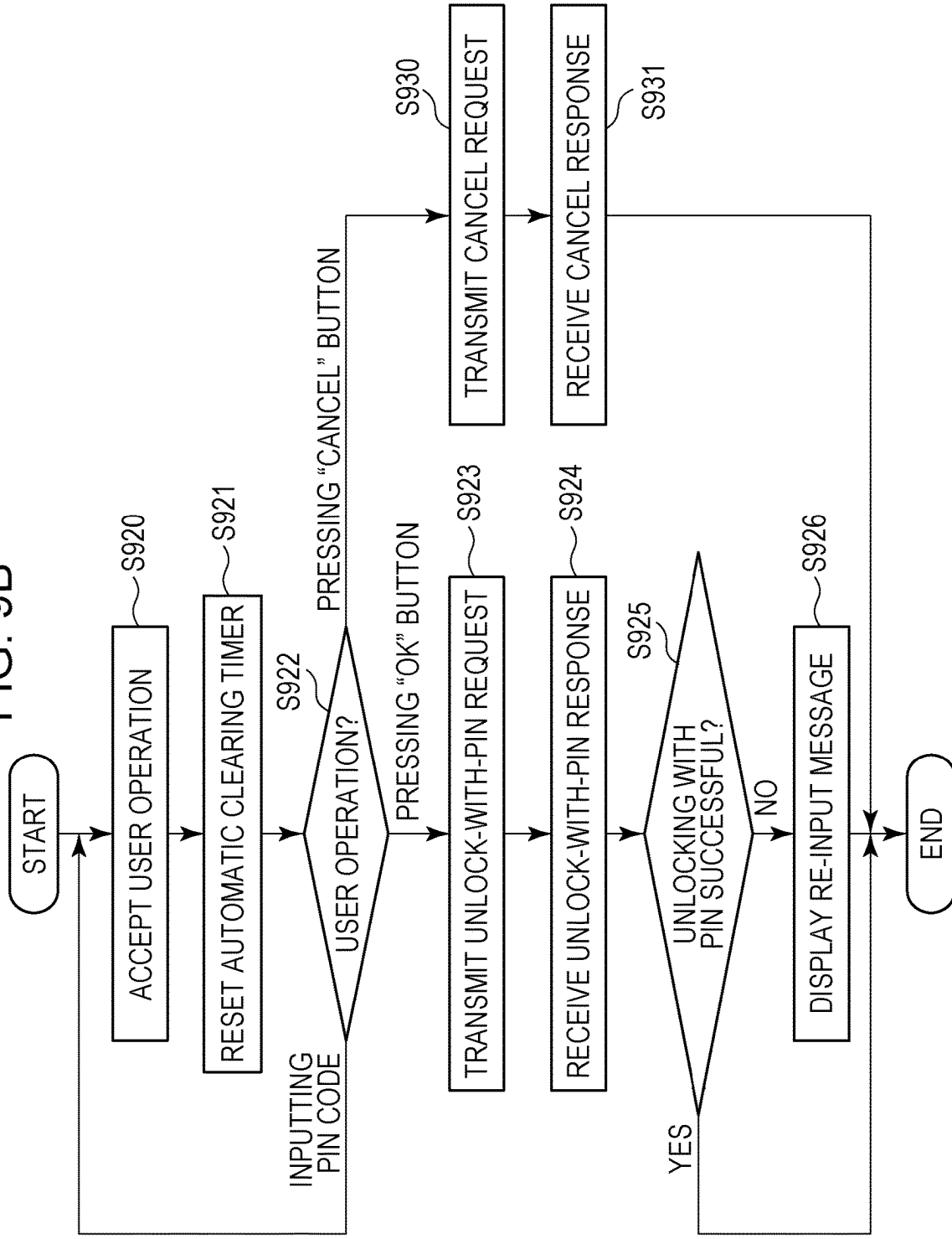


FIG. 10

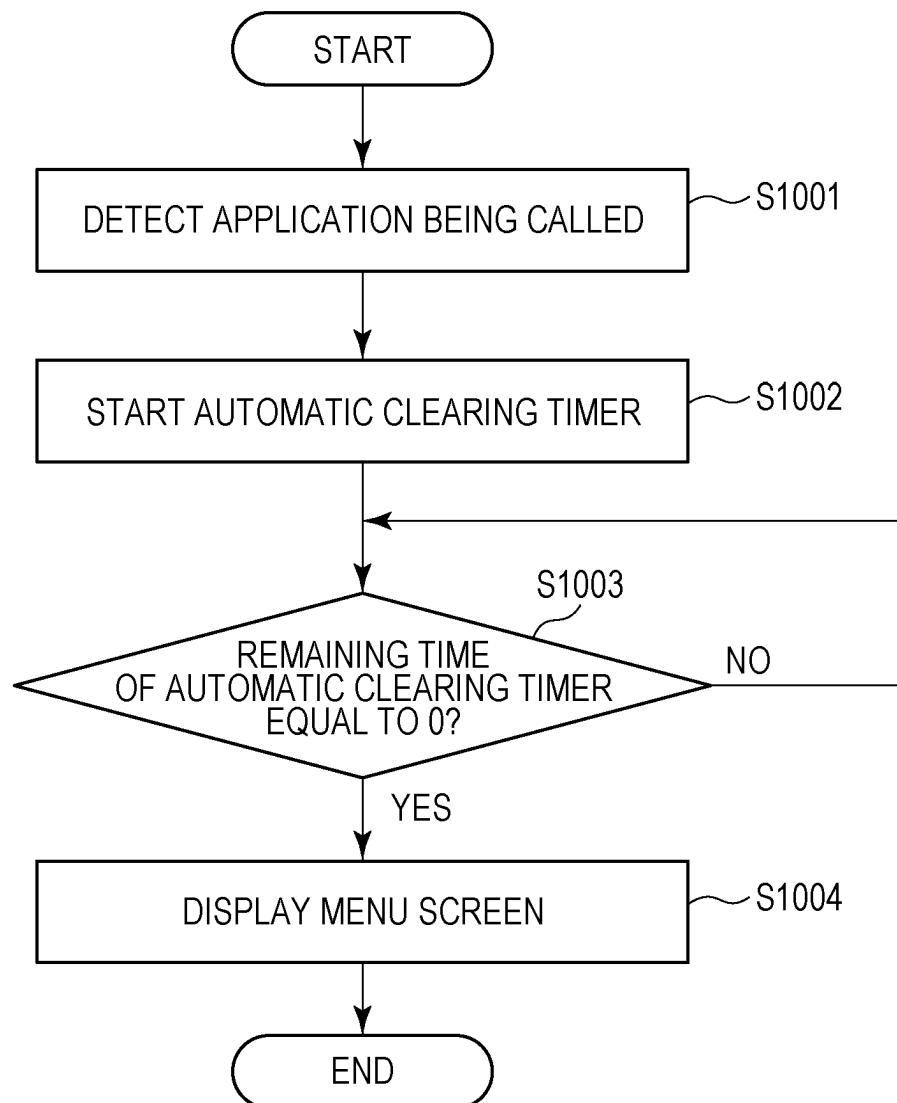


FIG. 11A

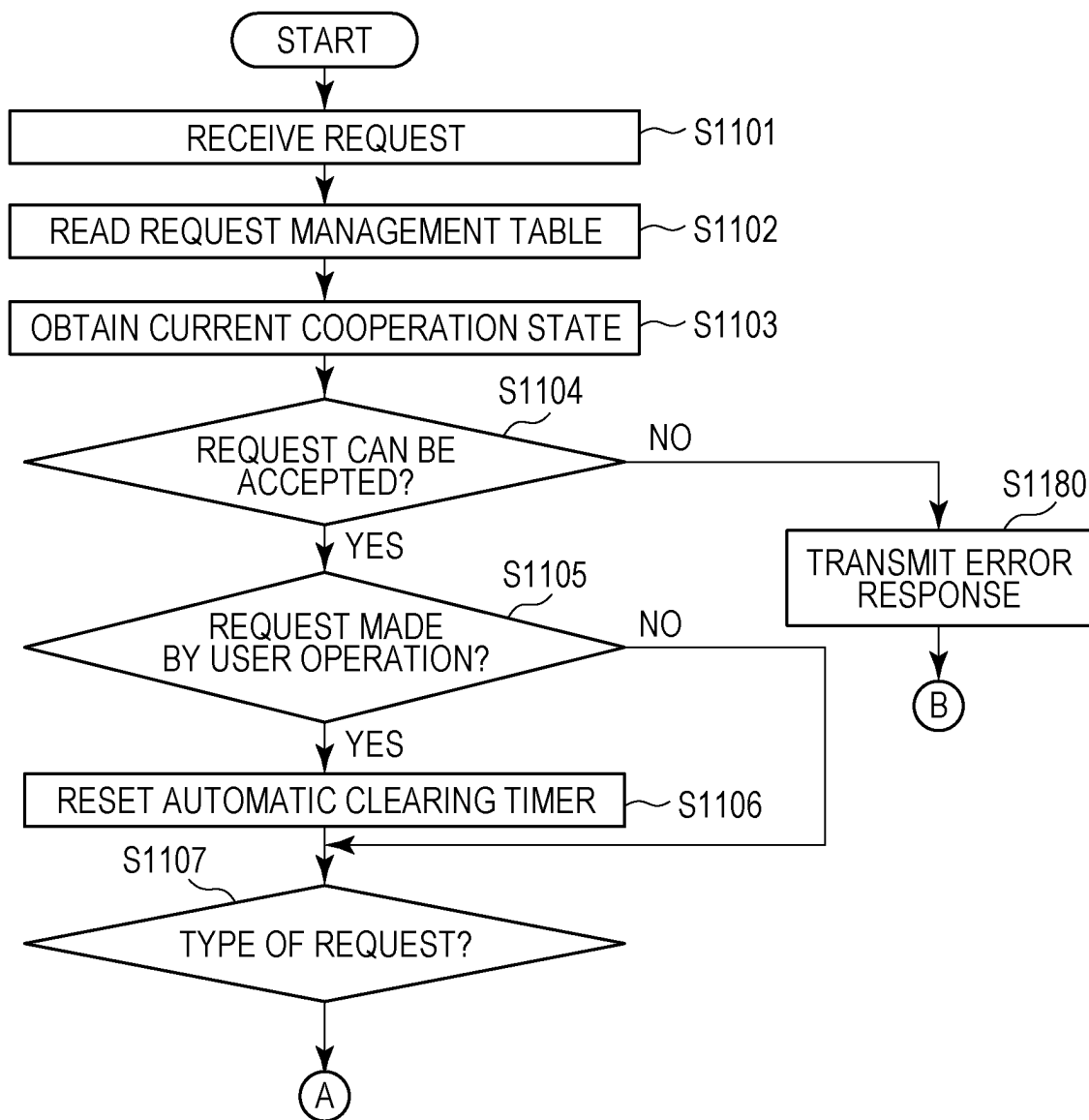


FIG. 11B

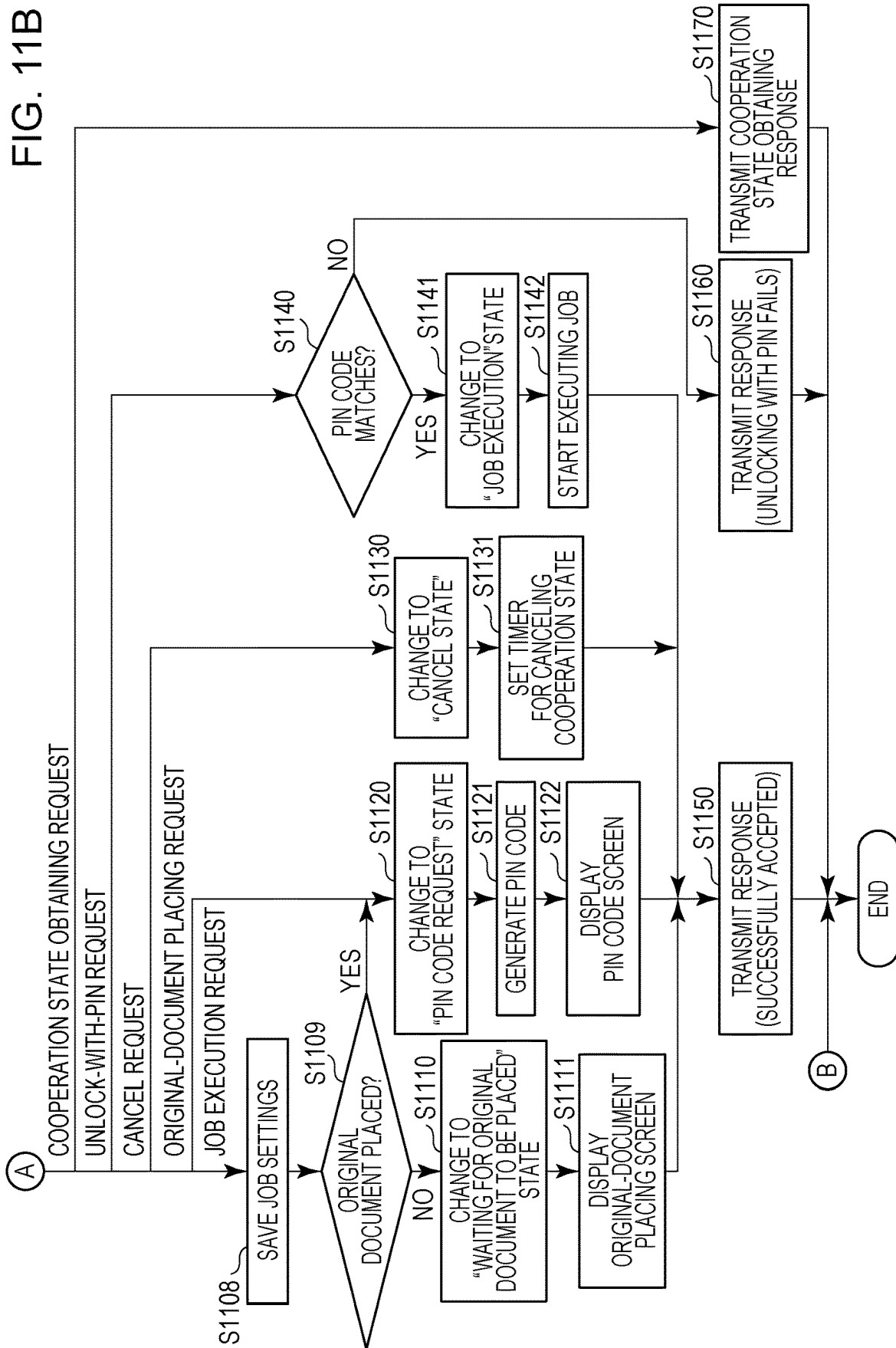


FIG. 12A

Enter your user name and password and press "Log in".

Make sure to log out at the end of your operation.

1202 User name

1203 Password

Logs in to:

1204 This device ▼

Log in

FIG. 12B

1211 Home

1210

1214

1213

Copy Facsimile Scan and transmit

Scan and save Use saving file Inbox

Save power Log out

Setting Home

Timeline

Stop

Check status

1212

FIG. 12C

Figure 1 is a schematic diagram of a document management system interface. The interface is divided into several sections. The top section, labeled 1221, contains "Scan and transmit" and "You can transmit." Below "You can transmit.", there are input fields for "Number of destinations: 2", "kato@example.com", and "yama@example.com", followed by left and right arrow buttons. The middle section, labeled 1222, contains "Save in my holder", "Specify destination", "Detail information", "Transmit to me", "Delete destination", and "CC BCC". The bottom section, labeled 1223, contains "History of settings" and "Frequently used settings". The "History of settings" list includes "Automatic", "300x300 dpi", "Automatic size", "PDF (High compression)", "Double-sided original document", "Original documents of different sizes mixed", and "Other functions". The "Frequently used settings" list includes "Automatic", "300x300 dpi", "Automatic size", "PDF (High compression)", "Double-sided original document", "Original documents of different sizes mixed", and "Other functions". The right side of the interface, labeled 1224, contains a "Save power" button, a "Setting/registration" button, and a "User log out" button. The bottom right corner, labeled 1225, contains a "Check status" button.

FIG. 12D

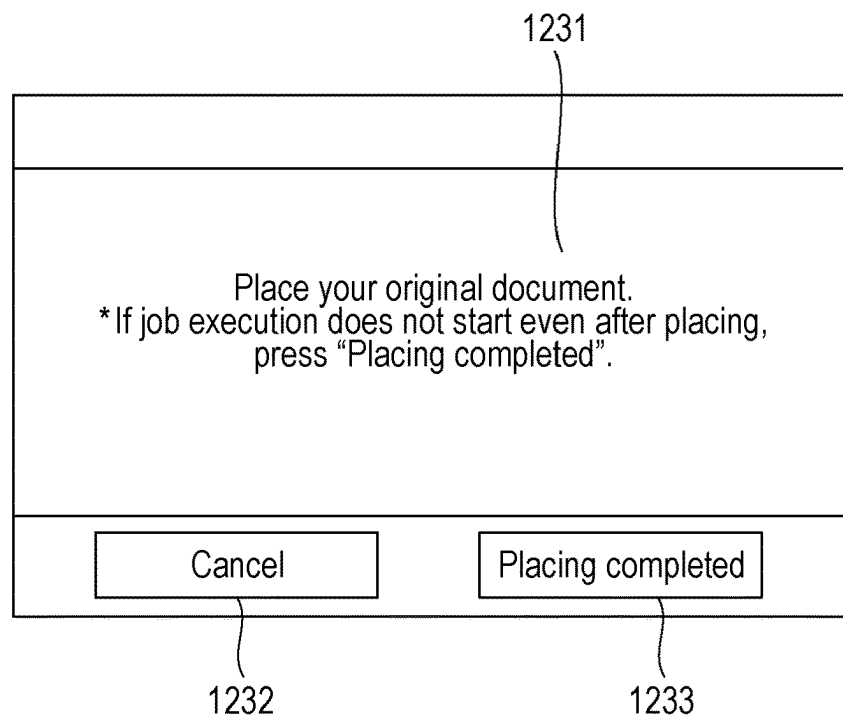


FIG. 12E

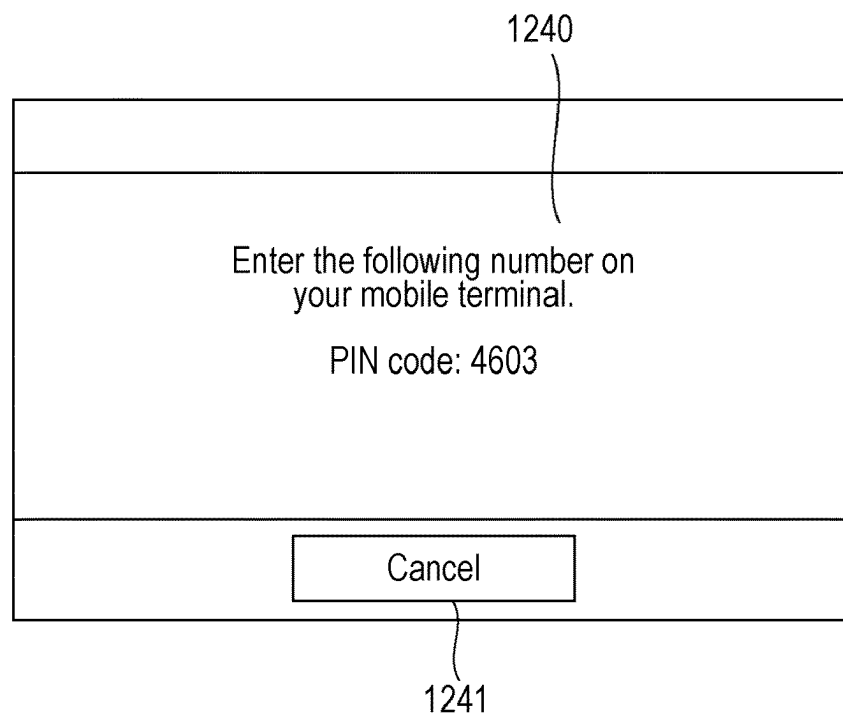


FIG. 13

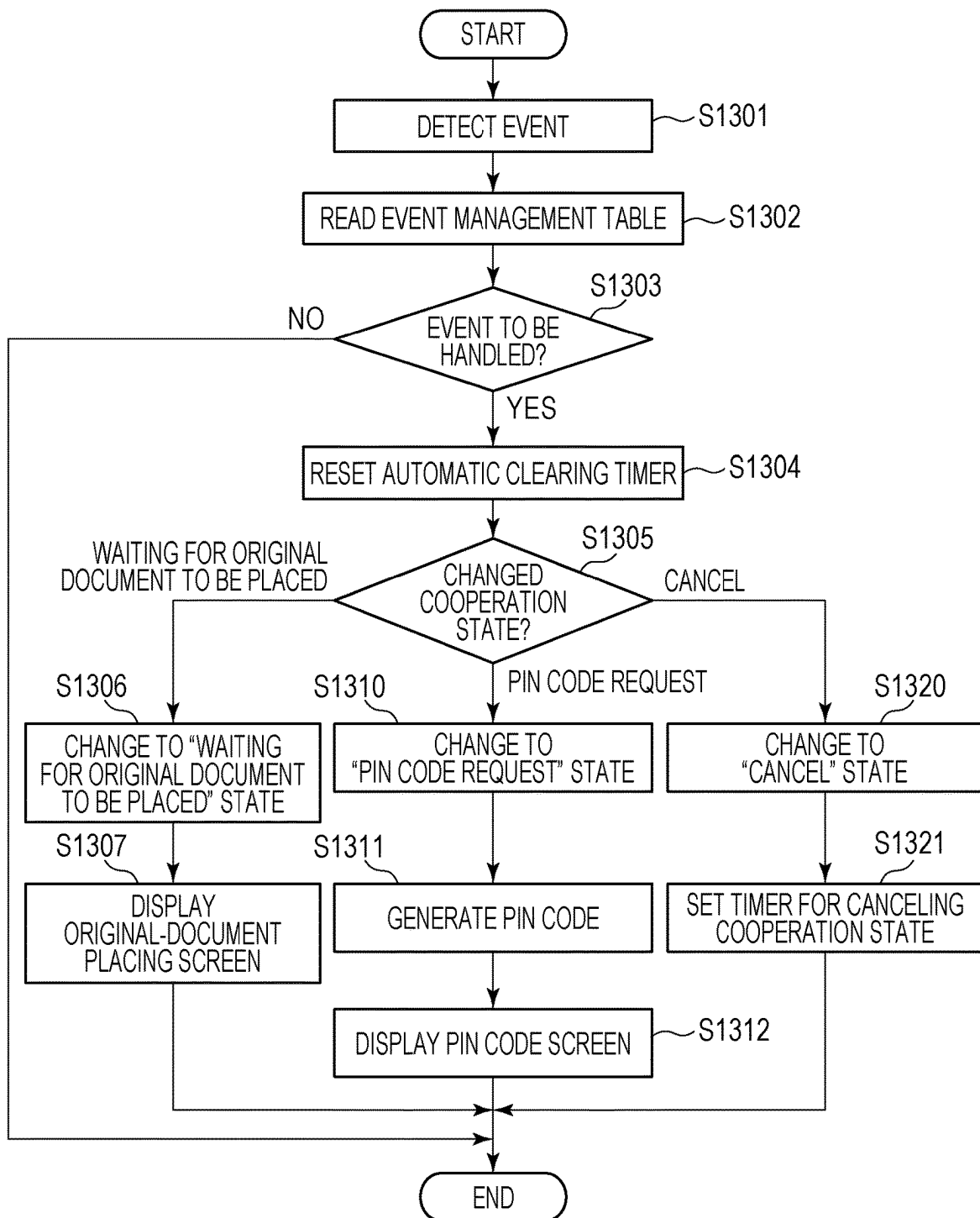


FIG. 14

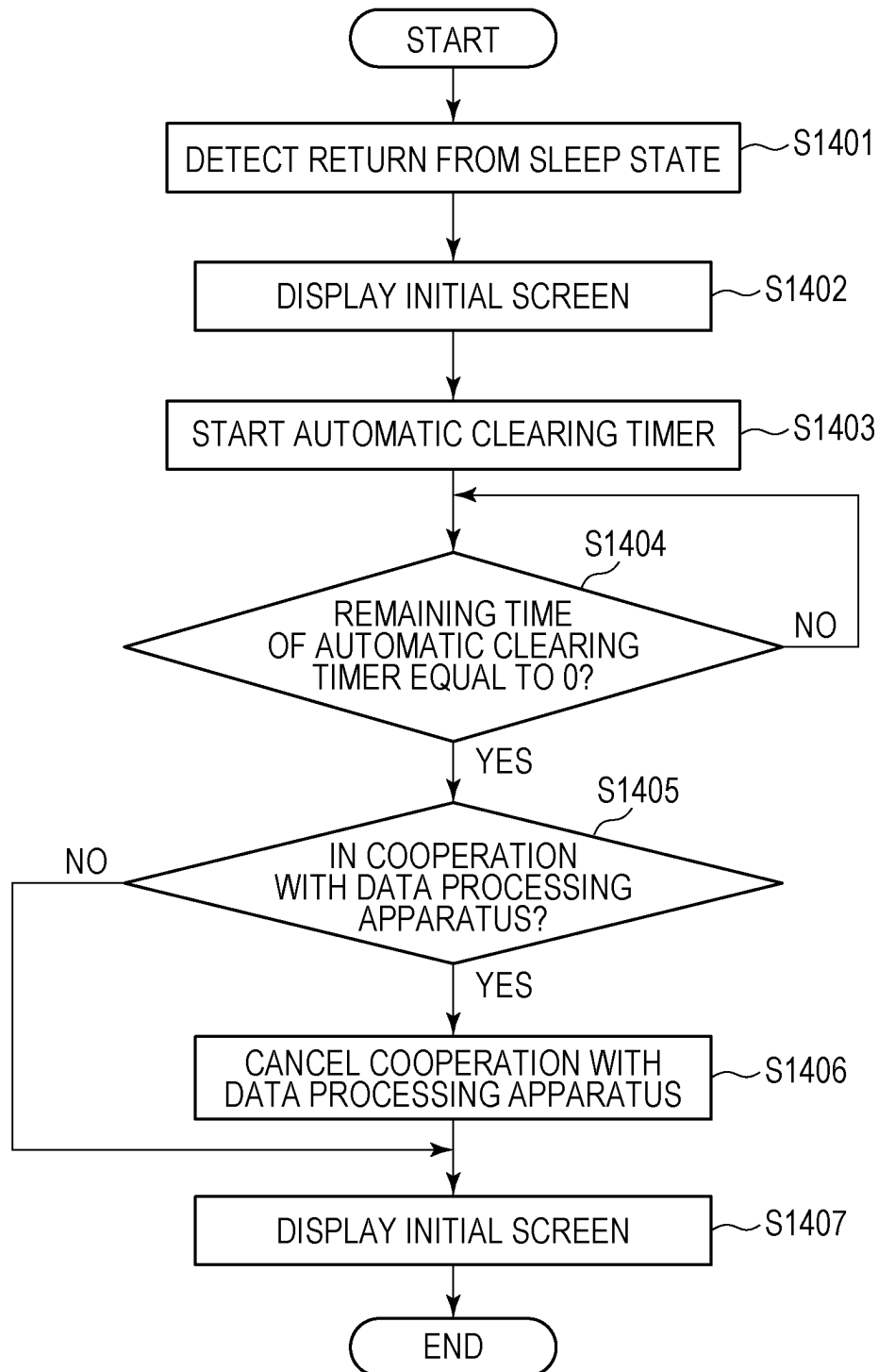


FIG. 15

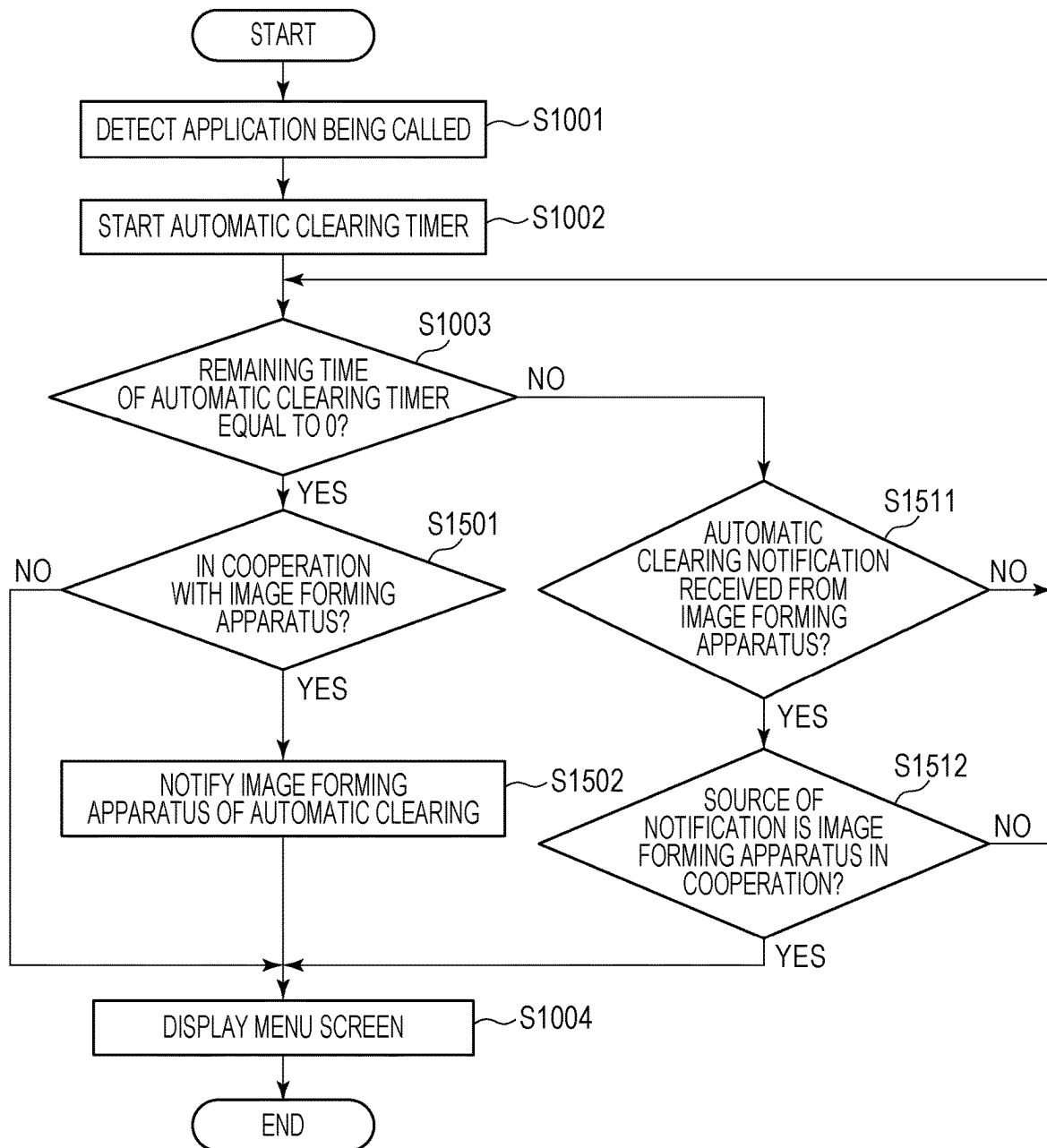


FIG. 16

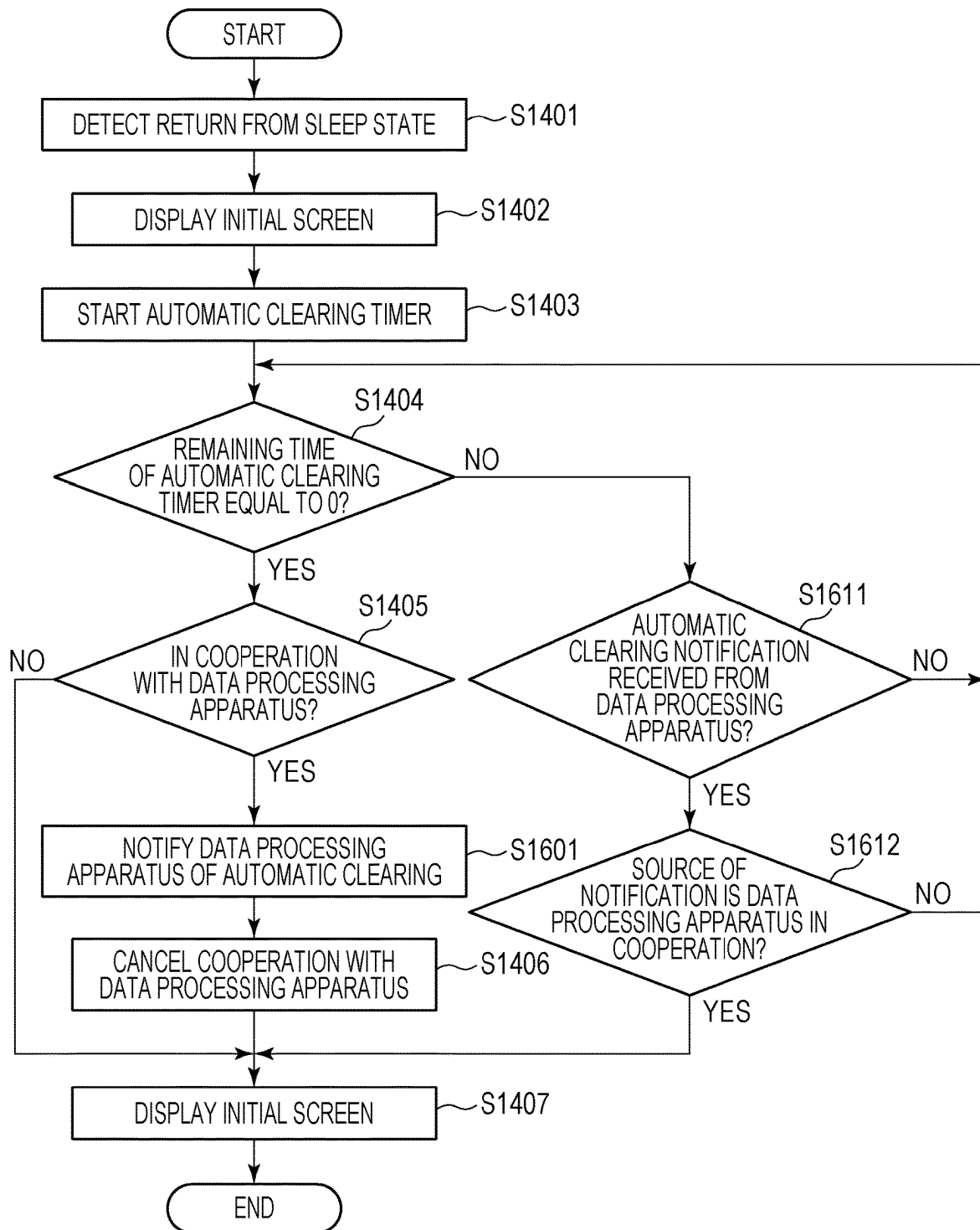
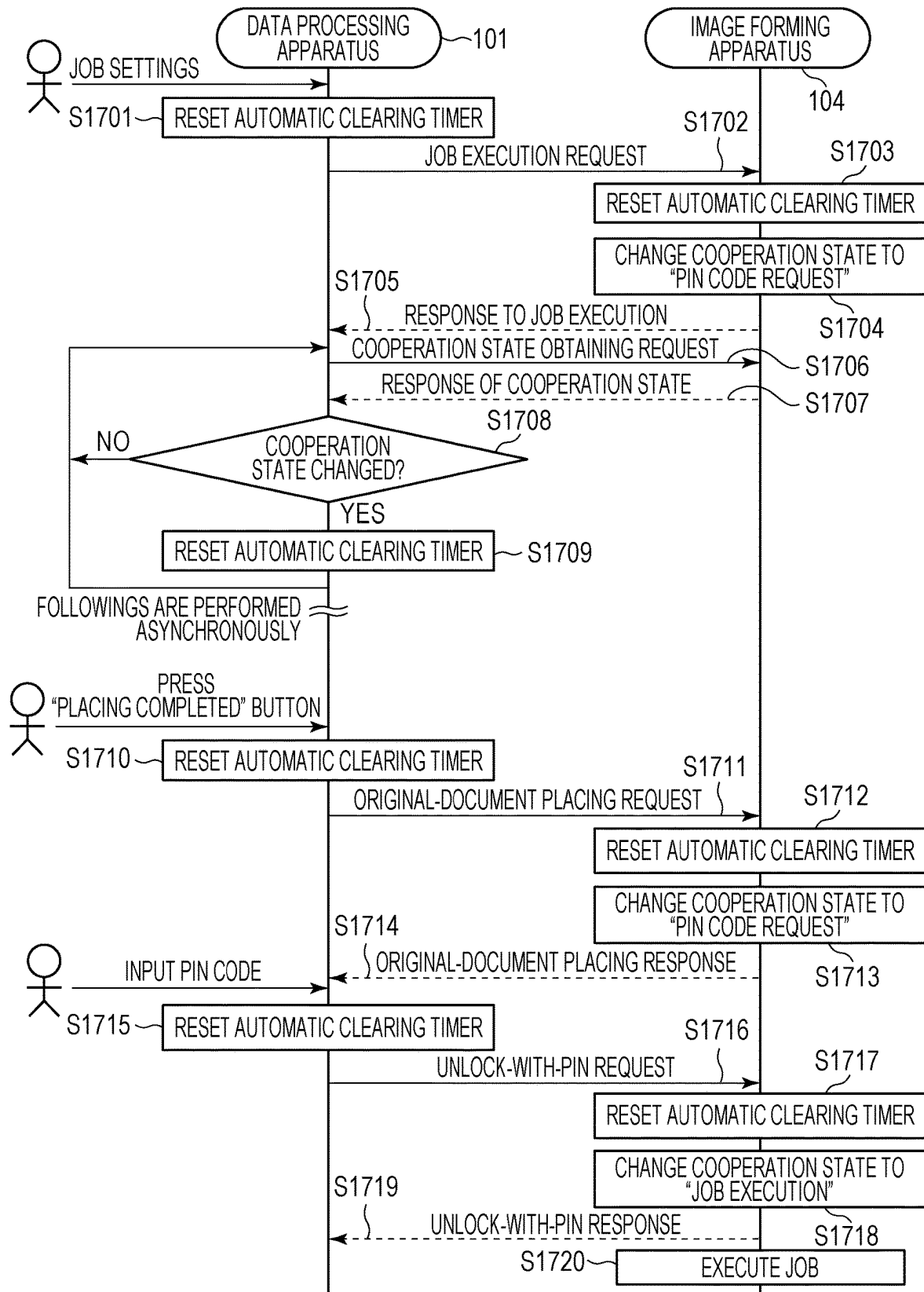


FIG. 17



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INFORMATION PROCESSING APPARATUS, INFORMATION PROCESSING SYSTEM, AND METHOD FOR INFORMATION PROCESSING APPARATUS

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a Continuation of U.S. patent application Ser. No. 17/936,768, filed Sep. 29, 2022, which claims the benefit of Japanese Patent Application No. 2021-161001, filed Sep. 30, 2021, both of which are hereby incorporated by reference herein in their entirety.

BACKGROUND OF THE INVENTION

Field of the Invention

The present invention relates to an information processing system in which apparatuses cooperate with each other to thereby execute a job.

Description of the Related Art

With the recent widespread use of high-performance mobile terminals called smartphones, applications for using an image forming apparatus from such mobile terminals have been provided. For example, to execute a function of transmitting a scanned original document to a specified destination, an image forming apparatus connected to a local area network (LAN) is searched for, and a job to which the email address of the destination and information about various scan settings (for example, color and double-sided) set by the user are added is generated. The generated job is transmitted to the image forming apparatus that has been found and that has a transmission function to thereby allow the mobile terminal to instruct the image forming apparatus to perform a series of processes necessary for transmission. At this time, network communication between the mobile terminal and the image forming apparatus is established in accordance with a communication protocol, such as Transmission Control Protocol/Internet Protocol (TCP/IP), Near Field Communication (NFC), or Bluetooth (registered trademark), and can be implemented with a publicly known technique.

A recent information processing apparatus, such as an image forming apparatus, has a function of clearing the set values of settings, restoring the display screen to an initial screen, or logging out in a case where no user operation on an operation unit is performed for a specific time. The function of restoring the apparatus to the initial state in a case where no operation is accepted for a specific time is defined as an automatic clearing function. Japanese Patent No. 3517183 proposes an image forming apparatus that can set the time (hereinafter referred to as "automatic clearing time") until the automatic clearing function is executed, on an operation mode by operation mode basis.

In a case where an information processing apparatus is operated while the information processing apparatus and a terminal are made to cooperate with each other, there is a problem that automatic clearing not intended by the user may occur, resulting in decreased usability. For example, a case is assumed where a job is transmitted from the terminal to the information processing apparatus and various processes necessary for executing the job are performed by operating the terminal. The user operates an operation unit of the terminal but does not operate an operation unit of the

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information processing apparatus, and therefore, an automatic clearing process is performed in the information processing apparatus after a predetermined time elapses.

SUMMARY OF THE INVENTION

The present invention provides a method for allowing an automatic clearing process to be appropriately performed even in a case where an information processing apparatus and a terminal are made to cooperate with each other.

The present invention provides an information processing apparatus configured to be capable of communicating with a terminal, the information processing apparatus including an operation unit, the information processing apparatus including: a counting unit configured to count a time during which no user operation on the operation unit is accepted; a performing unit configured to perform an automatic clearing process in a case where a predetermined time is counted by the counting unit; and a resetting unit configured to reset the time counted by the counting unit in response to a predetermined request received from the terminal.

Further features of the present invention will become apparent from the following description of embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a diagram illustrating a configuration of an information processing system.

FIG. 2 is a block diagram illustrating a hardware configuration of a data processing apparatus.

FIG. 3 is a block diagram illustrating a hardware configuration of an image forming apparatus.

FIG. 4 is a block diagram illustrating a software configuration of the data processing apparatus.

FIG. 5 is a block diagram illustrating a software configuration of the image forming apparatus.

FIGS. 6A and 6B illustrate an example of a cooperation management table of the image forming apparatus in a first embodiment.

FIG. 7 is a flowchart illustrating a process performed by the data processing apparatus in the first embodiment.

FIGS. 8A to 8E illustrate examples of screens of the data processing apparatus in the first embodiment.

FIGS. 9A and 9B are flowcharts illustrating processes performed by the data processing apparatus in the first embodiment.

FIG. 10 is a flowchart illustrating a process performed by the data processing apparatus in the first embodiment.

FIGS. 11A and 11B illustrate a flowchart illustrating a process performed by the image forming apparatus in the first embodiment.

FIGS. 12A to 12E illustrate examples of screens of the image forming apparatus in the first embodiment.

FIG. 13 is a flowchart illustrating a process performed by the image forming apparatus in the first embodiment.

FIG. 14 is a flowchart illustrating a process performed by the image forming apparatus in the first embodiment.

FIG. 15 is a flowchart illustrating a process performed by the data processing apparatus in a second embodiment.

FIG. 16 is a flowchart illustrating a process performed by the image forming apparatus in the second embodiment.

FIG. 17 is a sequence chart illustrating a process performed by the image forming apparatus and the data processing apparatus in the first embodiment.

DESCRIPTION OF THE EMBODIMENTS

Hereinafter, embodiments of the present invention will be described with reference to the drawings. Note that configu-

rations described in the embodiments are examples and are not intended to limit the scope of the present invention to the configurations. Although a description will be given below under the assumption that an image forming apparatus is an example of the information processing apparatus, the information processing apparatus is not limited to this.

First Embodiment

FIG. 1 is a diagram illustrating a configuration of an information processing system according to this embodiment. An information processing system 100 includes a data processing apparatus 101, a wireless LAN terminal 102, and an image forming apparatus 104, and these apparatuses are connected to each other over a network 103, which is, for example, a LAN. To the network 103, an apparatus other than those illustrated in FIG. 1 may be connected.

The data processing apparatus 101 is, for example, a mobile terminal, such as a smartphone, or may be an information processing apparatus on which an operating system for small terminals and programs for controlling voice calls, position detection, and data communication run. Alternatively, the data processing apparatus 101 may be a personal computer (PC) that does not have functions of voice call control, position detection control, and so on. The data processing apparatus 101 is connected to the network 103 via the wireless LAN terminal 102.

The wireless LAN terminal 102 is a wireless LAN base unit having a typical network router function and provides a wireless LAN, for example, at home or in an office.

The image forming apparatus 104 is a multifunction peripheral (MFP) having various functions including a printer function, a copy function, a scanner function, and a facsimile transmission function. However, the image forming apparatus 104 is not limited to this and may be an apparatus having a single function. The image forming apparatus 104 is configured to include a wireless access point so as to be capable of directly communicating with the data processing apparatus 101. A user can instruct the image forming apparatus 104 to execute jobs from the data processing apparatus 101 that communicates with the image forming apparatus 104 to thereby execute various functions of the image forming apparatus 104. Although the image forming apparatus 104 is configured to be connected to the network 103 by wire in this embodiment, the image forming apparatus 104 may be connected to the network 103 via the wireless LAN terminal 102 similarly to the data processing apparatus 101.

Further, the data processing apparatus 101 and the image forming apparatus 104 can perform short-range wireless communication conforming to, for example, NFC or Bluetooth Low Energy (BLE). The image forming apparatus 104 transmits to the data processing apparatus 101 via an NFC communication unit or a BLE communication unit described below, connection information (an Internet Protocol (IP) address or a Media Access Control (MAC) address, a Service Set Identifier (SSID), and so on) for a wireless LAN connection with the image forming apparatus 104. The data processing apparatus 101 uses the obtained connection information to start wireless LAN communication with the image forming apparatus 104.

FIG. 2 is a diagram illustrating a hardware configuration of the data processing apparatus 101 according to this embodiment. The data processing apparatus 101 includes a controller unit 201. The controller unit 201 controls various communication units including an NFC communication unit 210, a BLE communication unit 211, and a wireless LAN

communication unit 212 and various user interface (UI) units including a microphone-speaker 213, a display 214, and a touch panel 215.

The controller unit 201 includes a central processing unit (CPU) 202, a read-only memory (ROM) 203, a random access memory (RAM) 204, a network interface (UI) 205, an audio control unit 206, a display control unit 207, an input control unit 208, and a storage device 209, and these are connected to each other via a system bus 217.

The CPU 202 controls the entire system of the data processing apparatus 101. The ROM 203 stores the operating system of the data processing apparatus 101 and programs for controlling calls, data communication, and so on, and the CPU 202 executes the programs. The RAM 204 is used as a main memory of the CPU 202 and as a temporary storage area, such as a work area.

The storage device 209 is a nonvolatile storage device to which settings in various operation modes, an operation log, and so on that need to be retained even after the restart of the data processing apparatus 101 are recorded.

The network OF 205 is connected to the NFC communication unit 210, the BLE communication unit 211, and the wireless LAN communication unit 212 to exchange data with the image forming apparatus 104 through various types of wireless communication.

The audio control unit 206 controls input and output of audio data via the microphone-speaker 213. The display control unit 207 controls output of a screen to be displayed on the display 214. The input control unit 208 controls input of information specified by the user using a button or the touch panel 215. Various applications executed by the data processing apparatus 101 use the audio control unit 206, the display control unit 207, the input control unit 208, and so on.

FIG. 3 is a diagram illustrating a hardware configuration of the image forming apparatus 104 according to this embodiment. The image forming apparatus 104 includes a controller unit 301 and controls various communication units including an NFC communication unit 310, a BLE communication unit 311, and a wireless LAN communication unit 312, an operation unit 307, a scanner 313 that executes the scan function, and a printer 314 that executes the print function. In a case where a user uses the copy function of the image forming apparatus 104, the controller unit 301 controls the scanner 313 to obtain image data of an original document, and controls the printer 314 to print an image on a sheet and output the sheet. In a case where a user uses a "scan and transmit" function, the controller unit 301 controls the scanner 313 to obtain image data of an original document and convert the image data to code data. The controller unit 301 transmits the code data to, for example, the data processing apparatus 101 via a network I/F 308. In a case where a user uses the print function, the controller unit 301 receives image data (code data) from the data processing apparatus 101 via the network I/F 308. The controller unit 301 transmits the received image data to the printer 314. The printer 314 prints an image on a sheet and outputs the sheet on the basis of the received image data. The image forming apparatus 104 further has a facsimile reception function of receiving data from, for example, an Integrated Services Digital Network (ISDN) and printing the data and a facsimile transmission function of transmitting scanned data to, for example, an ISDN. To execute each of the functions as described above, the image forming apparatus 104 executes a job, which is a processing target task for which the user performs a setting operation and so on, in accordance with job information. For example, in a case of

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a “scan and transmit” job, a scan is performed in accordance with scan settings included in job information, and image data generated as a result of the scan is transmitted to a destination included in the job information.

The controller unit **301** includes a CPU **302**, a RAM **303**, a ROM **304**, a hard disk drive (HDD) **305**, an operation unit I/F **306**, the network I/F **308**, and a device I/F **309**, and these are connected to each other via a system bus **315**.

The CPU **302** controls the entire system of the image forming apparatus **104**. The RAM **303** is a system work memory for the CPU **302** to operate, and temporarily stores image data and so on.

The RAM **303** stores scan image data read by the scanner **313** and print image data received from the data processing apparatus **101** over the network **103**.

The ROM **304** stores a boot program for the system, applications, and so on. The HDD **305** stores an operating system and system software, application software, image data, setting data, and so on.

The operation unit **307** displays information for a user and accepts an operation from a user.

The operation unit **307** includes an operation panel, such as a touch panel display, and hardware keys.

The operation unit I/F **306** outputs to the operation unit **307** information to be displayed on the operation unit **307**.

The operation unit I/F **306** accepts input by a user from the operation unit **307** information. The network I/F **308** is connected to the NFC communication unit **310**, the BLE communication unit **311**, and the wireless LAN communication unit **312** to exchange information with the data processing apparatus **101** or a cloud server through various types of communication. The wireless LAN communication unit **312** performs wireless LAN communication with the data processing apparatus **101** over the network **103**. The NFC communication unit **310** and the BLE communication unit **311** perform short-range wireless communication with the data processing apparatus **101**. The image forming apparatus **104** receives setting information and image data of a job, a job execution command, and so on from the data processing apparatus **101** via the network I/F **308** and executes the job. The device I/F **309** connects the scanner **313** and the printer **314** that respectively read and print image data with the controller unit **301** to input and output image data.

FIG. **4** is a block diagram illustrating a software configuration of the data processing apparatus **101** according to this embodiment. FIG. **4** is a functional block diagram of software implemented by the CPU **202** reading a control program stored in the ROM **203** or the storage device **209**.

An application **401** is installed in the data processing apparatus **101** and is an application for operating the image forming apparatus **104**. The application **401** communicates with the image forming apparatus **104** and transmits various requests and receives various responses to and from the image forming apparatus **104** to thereby cooperate with the image forming apparatus **104**. For example, print job information or scan job information can be generated by performing a setting operation on the application **401**, and the job information can be transmitted to the image forming apparatus **104**. Although various applications are installed in the data processing apparatus **101** in addition to the application **401**, descriptions of such applications will be omitted. The configuration of the application **401** will be described below.

A UI control unit **402** accepts setting information of a job and connection information of the image forming apparatus **104** input by the user on the touch panel **215** and transmits

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the information to an MFP control unit **403**. The UI control unit **402** receives a response from the MFP control unit **403** and outputs the response to the display **214** via the display control unit **207**. The UI control unit **402** further controls the automatic clearing function as part of user interface control. Automatic clearing is a process of clearing the values of all settings set on a display screen or displaying an initial screen set in advance in a case where the data processing apparatus **101** accepts no operation from the user for a predetermined time. The initial screen can be set in advance by the user, and in a case where the user remains logged in to the data processing apparatus **101** upon the occurrence of automatic clearing, a log-out process is automatically performed. A count-down timer for measuring the remaining time until the occurrence of automatic clearing is hereinafter referred to as “automatic clearing timer”. The automatic clearing timer measures the time during which no operation is accepted from the user.

The MFP control unit **403** establishes communication with the image forming apparatus **104**, and transmits a request and receives a response to and from the image forming apparatus **104**. The MFP control unit **403** generates job execution instruction information from job settings accepted by the UI control unit **402** and transmits a job execution request to the image forming apparatus **104** over the network **103**. The MFP control unit **403** obtains a job execution status from the image forming apparatus **104** over the network **103** and transmits the job execution status to the UI control unit **402**.

FIG. **5** is a block diagram illustrating a software configuration of the image forming apparatus **104** according to this embodiment. System software **501** is software executed by the CPU **302** of the image forming apparatus **104** and is stored in the HDD **305**.

A UI control unit **502** accepts information input by a user on the operation unit **307** and, for example, controls a screen. The UI control unit **502** controls the automatic clearing function as part of user interface control. In a case where a user remains logged in to the image forming apparatus **104** upon the occurrence of automatic clearing, the UI control unit **502** automatically performs a log-out process.

A cooperation control unit **503** controls cooperation between the data processing apparatus **101** and the image forming apparatus **104**. Specifically, the cooperation control unit **503** controls the image forming apparatus **104** on the basis of a request received from the data processing apparatus **101** and transmits to the data processing apparatus **101** a response to the request. The cooperation control unit **503** manages the state (hereinafter referred to as “cooperation state”) of the image forming apparatus **104** when the data processing apparatus **101** and the image forming apparatus **104** cooperate with each other.

A cooperation management table **504** is a table for managing information necessary for the cooperation control unit **503** to control cooperation between the data processing apparatus **101** and the image forming apparatus **104**.

A job control unit **505** controls processing units including a scan processing unit **511**, a copy processing unit **512**, a print processing unit **513**, and a facsimile processing unit **514** in accordance with job settings and executes various jobs.

FIGS. **12A** to **12E** illustrate examples of screens displayed by the image forming apparatus **104** in this embodiment.

FIG. **12A** is a diagram illustrating an authentication screen of the image forming apparatus **104** according to this embodiment. A log-in screen **1201** is a screen for a user

authentication and includes a user name input field **1202** into which the user name of a user who locally logs in and a password input field **1203** into which a password is input. Local log-in is log-in for a user to exclusively operate the operation panel of the image forming apparatus **104**, and other users are not allowed to locally log in or operate the operation panel while the user remains locally logged in. When a log in button **1204** is pressed, the user name and the password respectively input into the user name input field **1202** and the password input field **1203** are used to start a local log-in process. Hereinafter, “log-in” refers to local log-in unless otherwise noted.

FIG. **12B** is a diagram illustrating a home screen of the image forming apparatus **104**, and the home screen is displayed when, for example, authentication on the log-in screen **1201** is successful and the user logs in to the image forming apparatus **104**.

A home screen **1210** is a screen for displaying functions of the image forming apparatus **104** in an array and is a screen for a user to select a function to be used. A copy button **1211** is a button for executing the copy function, and when pressed, a transition to a copy setting screen (not illustrated) takes place. When a check status button **1212** is pressed, a transition to a status check screen (not illustrated) takes place on which the user can refer to the execution status of a job and a history. A log out button **1213** displays the user name of a locally logged-in user, and when pressed, the logged-in user is made to log out and a transition to the log-in screen **1201** takes place. A “scan and transmit” button **1214** is a button for executing a function of transmitting a scanned image to a set destination. When the “scan and transmit” button **1214** is pressed, a transition to a “scan and transmit” setting screen **1220** takes place.

FIG. **12C** is a diagram illustrating a screen of the “scan and transmit” function of the image forming apparatus **104**. The “scan and transmit” setting screen **1220** is a screen for setting a scan and a destination of an image. In a destination field **1221**, already set destinations of images are displayed. A specify destination button **1222** is a button for setting a destination of an image, and when pressed, a list of destinations is displayed. A scan setting area **1223** is an area for performing a setting operation for a scan, and settings can be specified by pressing respective buttons. A start button **1224** is a button for performing a process on the basis of the specifics set on the “scan and transmit” setting screen **1220**, and when pressed, a scan is performed and an image generated as a result of the scan is transmitted to a set destination. When the start button **1224** is pressed, the image forming apparatus **104** displays a process-in-progress screen (not illustrated) to show the status of the process that is being performed. When the process ends, the “scan and transmit” setting screen **1220** is displayed, or a transition to the home screen **1210** takes place. A stop button **1225** is a button for stopping the process that is in progress.

FIG. **12D** illustrates an original-document placing screen for prompting a user to place an original document. An original-document placing screen **1231** is displayed in a case where an original document is not placed on the scanner of the image forming apparatus **104** upon reception of a “scan and transmit” job from the data processing apparatus **101**. When a cancel button **1232** is pressed, the job is canceled, and when an original document is placed or a placing completed button **1233** is pressed, the process starts.

FIG. **12E** is a Personal Identification Number (PIN) code screen for displaying a PIN code to be input on the data processing apparatus **101**. This screen is displayed upon the image forming apparatus **104** receiving a “scan and trans-

mit” job from the data processing apparatus **101**. To verify that the user of the data processing apparatus **101** is present in front of the image forming apparatus **104**, the user is made to input on the data processing apparatus **101** a PIN code displayed on a PIN code screen **1240**. The PIN code screen **1240** includes an area for displaying a PIN code generated in step **S1121** and a message for prompting the user to input the PIN code. A cancel button **1241** is a button for canceling the cooperation state. When a PIN code input on the data processing apparatus **101** is correct, the image forming apparatus **104** executes the “scan and transmit” job.

When the operation unit **307** accepts no operation from a user for a predetermined time while any of the screens illustrated in FIGS. **12B** to **12E** is displayed and an automatic clearing process is performed, the set values are cleared and the initial screen set in advance (for example, the home screen **1210**) is displayed. In a case where the user remains logged in to the image forming apparatus **104** upon the automatic clearing process, the user is made to log out and the log-in screen **1201** is displayed.

FIGS. **8A** to **8E** illustrate examples of screens that are displayed in a case where the application **401** of the data processing apparatus **101** performs a process for instructing the image forming apparatus **104** to execute a job.

FIG. **8A** illustrates an example of a job setting screen for “scan and transmit”. A job setting screen **811** is displayed when a “scan and transmit” button **852** is selected from a menu screen **851** of the application **401**. The job setting screen **811** allows a setting operation for transmitting by email, an image generated by scanning an original document by the image forming apparatus **104**. In a title **812**, a title is displayed and a button for canceling settings and returning to the previous screen is provided. In an email address setting area **813**, already set destinations are displayed, and when this area is pressed, a software keyboard for inputting an email address of a destination is displayed. When an address book start button **814** is pressed, an address book in the data processing apparatus **101** is started, and the email address of a destination can be selected. A plurality of email addresses can be set. When an open button **815** is pressed, switching to a display screen that allows setting of CC, BCC, the subject, the body, and the file name takes place, and these items can be input. In a scan setting area **816**, buttons for specifying scan settings including a color setting and a resolution when the image forming apparatus **104** performs a scan are provided.

When each of the buttons is pressed, a corresponding one of the settings can be changed. An execute job button **817** is a button for creating job information with the set values of the settings and transmitting the job information to the image forming apparatus **104**.

FIG. **8B** illustrates an original-document placing screen that is a screen displayed in a case where an original document is not placed on the image forming apparatus **104** upon transmission of “scan and transmit” job information to the image forming apparatus **104**. A message area **821** is an area for displaying a message for prompting the user to place an original document. A cancel button **822** is a button for canceling the cooperation state. A placing completed button **823** is a button for notifying the image forming apparatus **104** that an original document has been placed, in a case where the user has placed the original document on the scanner **313** of the image forming apparatus **104** but the image forming apparatus **104** (scanner **313**) does not detect the original document.

FIG. **8C** illustrates a PIN code input screen. This is a screen for inputting a PIN code included in the PIN code

screen **1240** displayed on the image forming apparatus **104** after transmission of “scan and transmit” job information to the image forming apparatus **104**. A PIN code input field **831** is an input field for inputting a PIN code. A cancel button **832** is a button for canceling the cooperation state. An OK button **833** is a button for transmitting a PIN code input into the PIN code input field **831** to the image forming apparatus **104**. A configuration may be employed in which a QR code (registered trademark) is displayed on the image forming apparatus **104** instead of a PIN code and the QR code is read by a camera **216** included in the data processing apparatus **101** to thereby transmit authentication information.

FIG. **8D** illustrates an end notification screen. This is a screen displayed when a PIN code input on the PIN code input screen is authenticated and job execution is started by the image forming apparatus **104**. A message area **841** is an area for displaying a message corresponding to a cooperation state received in step **S708**. In the message area **841** illustrated in FIG. **8D**, an example in a case where the cooperation state is “job execution” is displayed. An OK button **842** is a button for closing the end notification screen. When the end notification screen is closed, a transition to the menu screen of the application **401** takes place.

FIG. **8E** illustrates the menu screen of the application **401**. The menu screen **851** is the initial screen that is displayed when the application **401** is started, and the user selects a desired function from the menu screen. When the “scan and transmit” button **852** is pressed, the job setting screen **811** is displayed.

In a case where no operation is accepted from the user for a predetermined time while the data processing apparatus **101** displays any of the screens illustrated in FIGS. **8A** to **8D**, an automatic clearing process is performed, and the initial screen (for example, the menu screen illustrated in FIG. **8E**) is displayed.

As described above, when a “scan and transmit” job is set on the application **401** and job information is transmitted to the image forming apparatus **104**, and an original document is placed and a PIN code is input, the job is executed by the image forming apparatus **104**.

In a case where the data processing apparatus **101** is operated to make the image forming apparatus **104** perform a process, when the operation unit **307** of the image forming apparatus **104** accepts no operation from the user for a predetermined time and an automatic clearing process is performed as in the related art, a problem may arise. For example, when the image forming apparatus **104** determines that the operation unit **307** thereof accepts no operation even though the user performs an operation on the original-document placing screen illustrated in FIG. **8B** or the PIN code input screen illustrated in FIG. **8C** on the data processing apparatus **101**, keeps the automatic clearing timer counting, and performs an automatic clearing process, the original-document placing screen and the PIN code screen illustrated in FIG. **12E** are prevented from being displayed on the image forming apparatus **104** and, for example, the initial screen is displayed. Then, the user is, for example, unable to know a PIN code to be input and unable to make the process proceed. In this embodiment, when an operation is accepted by the data processing apparatus **101**, the image forming apparatus **104** is configured to reset the automatic clearing timer.

FIG. **17** is a sequence chart illustrating interaction between the image forming apparatus **104** and the data processing apparatus **101** in this embodiment.

When job settings from the user are accepted on the application **401** of the data processing apparatus **101** and an

instruction for transmitting job information to the image forming apparatus **104** is accepted, the data processing apparatus **101** resets the automatic clearing timer thereof in **S1701**, and a job execution request is transmitted from the data processing apparatus **101** to the image forming apparatus **104** in **S1702**. That is, job information for making the image forming apparatus **104** execute a job is transmitted to the image forming apparatus **104**. The remaining time of the automatic clearing timer decreases as time passes, and when the remaining time becomes equal to zero, automatic clearing occurs. When the automatic clearing timer is reset, the remaining time until the occurrence of automatic clearing is reset to the automatic clearing time. For example, in a case where the automatic clearing time is set to 120 seconds, the remaining time is reset to 120 seconds when the automatic clearing timer is reset.

When receiving the job execution request, the image forming apparatus **104** resets the automatic clearing timer thereof in **S1703**. In a case where an original document is placed on the image forming apparatus **104**, the cooperation state is changed to “PIN code request” in **S1704**, and a response is transmitted to the data processing apparatus **101** in **S1705**. The cooperation state will be described below.

The data processing apparatus **101** transmits a cooperation state obtaining request to the image forming apparatus **104** in **S1706**, and the image forming apparatus **104** returns a response in **S1707**. In **S1708**, the data processing apparatus **101** determines whether the cooperation state has changed on the basis of the received response information, and in a case where the cooperation state has changed, the data processing apparatus **101** resets the automatic clearing timer in **S1709** and performs display corresponding to the changed cooperation state. It is assumed that the request and response process in **S1706** and **S1707** is performed occasionally after transmission of the job information.

It is assumed that the process in **S1710** and the subsequent steps is performed asynchronously to the steps described above.

When pressing of the placing completed button **823** by the user is accepted on the application **401** of the data processing apparatus **101**, the data processing apparatus **101** resets the automatic clearing timer thereof in **S1710** and transmits an original-document placing request to the image forming apparatus **104** in **S1711**.

When receiving the original-document placing request, the image forming apparatus **104** resets the automatic clearing timer thereof in **S1712**. The image forming apparatus **104** changes the cooperation state to “PIN code request” in **S1713** and transmits an original-document placing response to the data processing apparatus **101** in **S1714**.

When accepting a PIN code input by the user on the application **401** of the data processing apparatus **101**, the data processing apparatus **101** resets the automatic clearing timer thereof in **S1715** and transmits an unlock-with-PIN request to the image forming apparatus **104** in **S1716**.

When receiving the unlock-with-PIN request, the image forming apparatus **104** resets the automatic clearing timer thereof in **S1717**. The image forming apparatus **104** changes the cooperation state to “job execution” in **S1718**, transmits an unlock-with-PIN response to the data processing apparatus **101** in **S1719**, and executes the job in **S1720** on the basis of the received job information.

FIGS. **6A** and **6B** illustrate an example of the cooperation management table **504** in this embodiment. The cooperation management table **504** includes two tables, namely a request management table **600** and an event management table **650**.

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FIG. 6A illustrates an example of the request management table **600** for managing requests received by the image forming apparatus **104** from the data processing apparatus **101**. One row defines information about one type of request.

A “request type” column **601** includes the type of a request received by the image forming apparatus **104**.

A “job execution request” is a request for instructing the image forming apparatus **104** to execute a job. A “cooperation state obtaining request” is a request for obtaining state information from the image forming apparatus **104**. An “original-document placing request” is a request for a notification that an original document has been placed. An “unlock-with-PIN request” is a request for performing authentication with an input PIN code. A “cancel request” is a request for canceling execution of a job.

A “cooperation state for acceptance” column **602** defines a cooperation state in which the request included in the “request type” column **601**, when received, can be accepted. The table shows, for example, that the “job execution request” can be accepted only in a case where the cooperation state is “(not in cooperation)”. The cooperation state “(not in cooperation)” refers to a state in which, for example, the data processing apparatus **101** and the image forming apparatus **104** communicate with each other but job information is not transmitted from the data processing apparatus **101** to the image forming apparatus **104**, and a cooperation state “(in cooperation)” refers to a state in which job information has been transmitted from the data processing apparatus **101** to the image forming apparatus **104** and the apparatuses can cooperate with each other. As described above, the cooperation state indicates a stage or a status in a process performed by exchanging information between the data processing apparatus **101** and the image forming apparatus **104**. The “cooperation state obtaining request” can be accepted only in a case where a job has been transmitted from the data processing apparatus **101** to the image forming apparatus **104** and the cooperation state is “(in cooperation)”.

A “user-operation-causing request” column **603** defines whether the request in the “request type” column **601** is a request caused by a user operation. In a case of a row in which the “user-operation-causing request” column **603** includes “YES”, the request is a request caused by a user operation. In a case of a row in which the “user-operation-causing request” column **603** includes “NO”, the request is a request transmitted regardless of a user operation. For example, the “job execution request” is a request that is transmitted when the user presses a button for transmitting a job on the data processing apparatus **101**, and therefore, “YES” is set in the “user-operation-causing request” column **603**. The “cooperation state obtaining request” is not a request that is transmitted when the user operates the data processing apparatus **101**, and therefore, “NO” is set in the “user-operation-causing request” column **603**.

FIG. 6B illustrates an example of the event management table **650** for managing event information about an event based on which the cooperation state changes. One row defines information about one cooperation state, event information about an event that is handled in the cooperation state, and a changed cooperation state after the occurrence of the event. A “cooperation state” column **651** includes one cooperation state. An “occurring event” column **652** defines an event to be handled when the image forming apparatus **104** is in the state included in the “cooperation state” column **651**, and the cooperation state is not changed in a case of the occurrence of an event not defined in the “occurring event” column **652**.

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A “changed cooperation state” column **653** defines a cooperation state after the occurrence of the event defined in the “occurring event” column **652**. A description of, for example, a row **662** will be given. In a case where either a “scanner: an original document placed” event or a “UI: the placing completed button pressed” event occurs while the cooperation state is “waiting for an original document to be placed”, the cooperation state transitions to “PIN code request”.

With reference to FIG. 7 and FIGS. 8A to 8E, a process for a job execution instruction performed by the data processing apparatus **101** of this embodiment will be described. FIG. 7 is a flowchart for explaining the process for a job execution instruction, and the process is implemented by the CPU **202** of the data processing apparatus **101** loading a program for implementing a control module stored in the ROM **203** or the storage device **209** to the RAM **204** and executing the program.

In step **S701**, the UI control unit **402** accepts information (an IP address and so on) for a connection with the image forming apparatus **104** that executes a job, the information being input by the user performing an operation on a screen of the application **401**. The information for a connection with the image forming apparatus **104** need not be input by the user but may be obtained through short-range wireless communication, such as NFC. The wireless LAN communication unit **212** may be used to search for an image forming apparatus that is connected to the network **103**. It is assumed that the information accepted in step **S701** is used to communicate with the image forming apparatus **104** in the subsequent steps.

In step **S702**, the UI control unit **402** accepts job settings and a transmission instruction for job information input by the user performing an operation on the job setting screen **811**. Specifically, scan settings and a destination setting are accepted on the job setting screen **811**, and pressing of the execute job button **817** is accepted.

In step **S703**, the UI control unit **402** resets the automatic clearing timer of the data processing apparatus **101**.

In step **S704**, the MFP control unit **403** transmits a job execution request that includes the job information created on the basis of the operation accepted in step **S702** to the image forming apparatus **104**.

In step **S705**, the MFP control unit **403** receives a job execution response from the image forming apparatus **104** as a response to the request transmitted in step **S704**. In step **S706**, the MFP control unit **403** determines the content of the job execution response. If the content of the job execution response indicates a success, the MFP control unit **403** makes the process proceed to step **S707**; otherwise the MFP control unit **403** makes the process proceed to step **S750**.

In step **S707**, the MFP control unit **403** transmits a cooperation state obtaining request for obtaining the cooperation state to the image forming apparatus **104**. In step **S708**, the MFP control unit **403** receives a cooperation state obtaining response from the image forming apparatus **104** as a response to the request transmitted in step **S707**. The cooperation state obtaining response includes the cooperation state of the image forming apparatus **104** and other information.

In step **S709**, the MFP control unit **403** determines whether the cooperation state received in step **S708** has changed from the immediately preceding cooperation state. If the cooperation state has changed, the UI control unit **402** makes the process proceed to step **S710**; otherwise the UI control unit **402** makes the process return to step **S707**.

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In step S710, the UI control unit 402 resets the automatic clearing timer of the data processing apparatus 101 in response to the changed cooperation state. Although a change in the cooperation state is used as an example of the basis for determining whether to reset the automatic clearing timer upon reception of the cooperation state obtaining response in this flowchart, another basis for determination may be used. For example, determination may be performed on the basis of the type of the cooperation state included in the cooperation state obtaining response or determination may be performed by using other information (for example, information about whether a user operation is performed on the image forming apparatus 104) included in the cooperation state obtaining response.

In step S711, the MFP control unit 403 makes the process proceed to step S712 if the cooperation state received in step S708 is "waiting for an original document to be placed", makes the process proceed to step S720 if the cooperation state received in step S708 is "PIN code request", or makes the process proceed to step S730 if the cooperation state received in step S708 is "job execution" or "cancel".

In step S712, the UI control unit 402 displays the original-document placing screen (FIG. 8B) on the display 214, and subsequently, makes the process return to step S707. The original-document placing screen is a screen for displaying a message for prompting the user to place an original document on the scanner 313 of the image forming apparatus 104. A process that is performed in a case where a user operation is performed on the original-document placing screen will be described below with reference to the flowchart in FIG. 9A.

In step S720, the UI control unit 402 displays the PIN code input screen (FIG. 8C) on the display 214, and subsequently, makes the process return to step S707. The PIN code input screen is a screen for inputting a PIN code displayed on the operation unit 307 of the image forming apparatus 104. A process that is performed in a case where a user operation is performed on the PIN code input screen will be described below with reference to the flowchart in FIG. 9B.

In step S730, the UI control unit 402 displays the end notification screen (FIG. 8D) on the display 214 and ends the process in the flowchart. The end notification screen is a screen for notifying the user of the end state when the cooperation state with the image forming apparatus 104 ends.

In step S750, the UI control unit 402 displays on the display 214 a message saying that the job execution instruction results in an error and ends the process in the flowchart.

With the above-described process in the flowchart, even if the operation unit of the data processing apparatus 101 accepts no operation from the user for a predetermined time during cooperation between the data processing apparatus 101 and the image forming apparatus 104, an automatic clearing process can be prevented from being performed in the data processing apparatus 101.

With reference to FIGS. 9A and 9B, processes performed in a case where user operations are performed on the application 401 will be described. FIGS. 9A and 9B are flowcharts for explaining processes performed in the data processing apparatus 101, and the processes are implemented by the CPU 202 of the data processing apparatus 101 loading a program for implementing a control module stored in the ROM 203 or the storage device 209 to the RAM 204 and executing the program. The processes in the flowcharts illustrated in FIGS. 9A and 9B are performed asynchronously to the process in the flowchart illustrated in FIG. 7.

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FIG. 9A is a flowchart for explaining a process performed in a case where a user operation is performed on the original-document placing screen (FIG. 8B) displayed in step S712.

In step S901, the UI control unit 402 accepts a user operation performed on the original-document placing screen and the process proceeds to the next step.

In step S902, the UI control unit 402 resets the automatic clearing timer of the data processing apparatus 101. In step S903, the UI control unit 402 determines the user operation detected in step S901. The UI control unit 402 makes the process proceed to step S904 if the user operation is pressing of the placing completed button 823, or makes the process proceed to step S910 if the user operation is pressing of the cancel button 822.

In step S904, the MFP control unit 403 transmits to the image forming apparatus 104 an original-document placing request for giving a notification that an original document has been placed. In step S905, the MFP control unit 403 receives from the image forming apparatus 104 an original-document placing response as a response to the request transmitted in step S904 and ends the process in the flowchart.

In step S910, the MFP control unit 403 transmits to the image forming apparatus 104 a cancel request for canceling the cooperation state. In step S911, the MFP control unit 403 receives a cancel response from the image forming apparatus 104 as a response to the request transmitted in step S910 and ends the process in the flowchart.

FIG. 9B is a flowchart for explaining a process performed in a case where a user operation is performed on the PIN code input screen (FIG. 8C) displayed in step S720.

In step S920, the UI control unit 402 accepts a user operation performed on the PIN code input screen, and the process proceeds to the next step.

In step S921, the UI control unit 402 resets the automatic clearing timer of the data processing apparatus 101. In step S922, the UI control unit 402 determines the user operation accepted in step S920. The UI control unit 402 makes the process return to step S920 if the user operation is input of a PIN code, makes the process proceed to step S923 if the user operation is pressing of the OK button 833, or makes the process proceed to step S930 if the user operation is pressing of the cancel button 832.

In step S923, the MFP control unit 403 transmits to the image forming apparatus 104 an unlock-with-PIN request for unlocking with the PIN code. The unlock-with-PIN request includes the PIN code input into the PIN code input field 831 by the user. In step S924, the MFP control unit 403 receives an unlock-with-PIN response from the image forming apparatus 104 as a response to the request transmitted in step S923. The unlock-with-PIN response includes information about whether unlocking with the PIN code is successful. In step S925, the MFP control unit 403 ends the process in the flowchart if unlocking with the PIN code is successful or makes the process proceed to step S926 if unlocking with the PIN code fails. In step S926, the UI control unit 402 displays a message saying that unlocking with the PIN code fails and prompting the user to re-input a PIN code into a message area (PIN code input field 831), and ends the process in the flowchart.

In step S930, the MFP control unit 403 transmits to the image forming apparatus 104 a cancel request for canceling the cooperation state. In step S931, the MFP control unit 403 receives a cancel response from the image forming apparatus 104 as a response to the request transmitted in step S930 and ends the process in the flowchart.

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With reference to FIG. 10, a process for automatic clearing control in the data processing apparatus 101 will be described. FIG. 10 is a flowchart for explaining a process performed by the data processing apparatus 101, and the process is implemented by the CPU 202 of the data processing apparatus 101 loading a program for implementing a control module stored in the ROM 203 or the storage device 209 to the RAM 204 and executing the program. The process in the flowchart illustrated in FIG. 10 is performed asynchronously to the process in the flowchart illustrated in FIG. 7 and to the processes in the flowcharts illustrated in FIGS. 9A and 9B.

Note that in this embodiment, an example case where the menu screen of the application 401 is displayed in response to the occurrence of automatic clearing in the data processing apparatus 101 will be described. However, this embodiment is not limited to this, and the home screen of the data processing apparatus 101 may be displayed or the data processing apparatus 101 may be locked in a case where the data processing apparatus 101 is a mobile terminal.

In step S1001, the UI control unit 402 detects the application 401 being called and displayed on the display 214. In step S1002, the UI control unit 402 starts counting down by the automatic clearing timer. Thereafter, the automatic clearing timer decreases the remaining time as time passes.

In step S1003, the UI control unit 402 monitors the remaining time of the automatic clearing timer, and makes the process proceed to step S1004 when detecting the remaining time being equal to zero or keeps monitoring the remaining time if the remaining time is more than zero.

In step S1004, the UI control unit 402 displays the menu screen 851 of the data processing apparatus 101 and ends the process in the flowchart.

With the process in the flowchart described above, an automatic clearing process is performed when a predetermined time elapses in a state in which a predetermined condition is satisfied. Although an example of counting down has been described here, counting up may be performed and automatic clearing may be made to occur when the count reaches a predetermined count.

With reference to FIGS. 11A and 11B and FIG. 12, a cooperation process that is performed by the image forming apparatus 104 of this embodiment with the data processing apparatus 101 will be described. FIGS. 11A and 11B illustrate a flowchart for explaining the cooperation process in the image forming apparatus 104, and the process is implemented by the CPU 302 of the image forming apparatus 104 loading a program for implementing a control module stored in the ROM 304 or the HDD 305 to the RAM 303 and executing the program.

In step S1101, the cooperation control unit 503 receives over the network 103 a request transmitted from the data processing apparatus 101. This request is, for example, information transmitted in step S704 or step S707.

In step S1102, the cooperation control unit 503 obtains from the request management table 600 information (information defined in the “cooperation state for acceptance” column 602 and in the “user-operation-causing request” column 603) corresponding to the request received in step S1101.

In step S1103, the cooperation control unit 503 obtains the current cooperation state of the image forming apparatus 104. In step S1104, the cooperation control unit 503 determines whether the request received in step S1101 is a request that can be accepted in the current cooperation state. Specifically, the cooperation control unit 503 determines whether the current cooperation state satisfies the value in

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the “cooperation state for acceptance” column 602 obtained in step S1102. If the request is a request that can be accepted, the cooperation control unit 503 makes the process proceed to step S1105; otherwise the cooperation control unit 503 makes the process proceed to step S1180.

In step S1105, the cooperation control unit 503 determines whether the request received in step S1101 is a request made by a user operation. Specifically, the cooperation control unit 503 performs determination on the basis of whether the value in the “user-operation-causing request” column 603 obtained in step S1102 is “YES” or “NO”. Performing determination based on the value in the “user-operation-causing request” column 603 as to whether the request is a request made by a user operation is an example, and other methods may be used. For example, a request transmitted by the data processing apparatus 101 may include information indicating whether the request is a request made by a user operation, and determination may be performed on the basis of the information. If the request is a request made by a user operation, the cooperation control unit 503 makes the process proceed to step S1106; otherwise the cooperation control unit 503 makes the process proceed to step S1107.

In step S1106, the UI control unit 502 resets the automatic clearing timer of the image forming apparatus 104.

In step S1107, the cooperation control unit 503 makes the process diverge in accordance with the type of the request received in step S1101. If the request is a “job execution request”, the cooperation control unit 503 makes the process proceed to step S1108. If the request is an “original-document placing request”, the cooperation control unit 503 makes the process proceed to step S1120. If the request is a “cancel request”, the cooperation control unit 503 makes the process proceed to step S1130. If the request is an “unlock-with-PIN request”, the cooperation control unit 503 makes the process proceed to step S1140. If the request is a “cooperation state obtaining request”, the cooperation control unit 503 makes the process proceed to step S1170.

In step S1108, the cooperation control unit 503 saves job information included in the job execution request received in step S1101 in the RAM 303. The cooperation control unit 503 retains the saved job information until the cooperation state is canceled. In step S1109, the cooperation control unit 503 determines the original-document detection state of the scanner 313 of the image forming apparatus 104. If the original-document detection state of the scanner 313 is a state in which a placed original document is detected, the cooperation control unit 503 makes the process proceed to step S1120; otherwise the cooperation control unit 503 makes the process proceed to step S1110.

In step S1110, the cooperation control unit 503 changes the cooperation state to the “waiting for an original document to be placed” state.

In step S1111, the UI control unit 502 displays the original-document placing screen (FIG. 12D) on the operation unit 307 of the image forming apparatus 104, and subsequently, makes the process proceed to step S1150. The original-document placing screen is a screen for displaying a message for prompting the user to place an original document on the scanner 313. A process that is performed in a case where a user operation is performed on the original-document placing screen will be described with reference to the flowchart in FIG. 13.

In step S1120, the cooperation control unit 503 changes the cooperation state to the “PIN code request” state. In step S1121, the cooperation control unit 503 generates a PIN code. Although the PIN code may be generated in any

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manner, in this embodiment, for example, a four-digit number generated at random may be used as the PIN code. In step S1122, the UI control unit 502 displays the PIN code screen (FIG. 12E) on the operation unit 307 of the image forming apparatus 104, and subsequently, makes the process proceed to step S1150. The PIN code screen is a screen for presenting to the user the PIN code generated in step S1121. A process that is performed in a case where a user operation is performed on the PIN code screen will be described with reference to the flowchart in FIG. 13.

In step S1130, the cooperation control unit 503 changes the cooperation state to the “cancel” state. In step S1131, the cooperation control unit 503 sets a timer for canceling the cooperation state, and subsequently, makes the process proceed to step S1150. After the elapse of the time set for the timer, the cooperation control unit 503 deletes the cooperation state and the job settings saved in step S1108, and the image forming apparatus 104 transitions to the “not in cooperation” state.

In step S1140, the cooperation control unit 503 determines whether a PIN code included in the unlock-with-PIN request received in step S1101 matches with the PIN code generated in step S1121. If the PIN codes match, the cooperation control unit 503 makes the process proceed to step S1141; otherwise the cooperation control unit 503 makes the process proceed to step S1160.

In step S1141, the cooperation control unit 503 changes the cooperation state to the “job execution” state. In step S1142, the cooperation control unit 503 requests the job control unit 505 to start executing the job by using the job settings saved in step S1108. Subsequently, the process proceeds to step S1150.

In step S1150, the cooperation control unit 503 transmits to the data processing apparatus 101 a response to the request received in step S1101 and ends the process in the flowchart. The response includes information indicating that the request is successfully accepted. For example, in a case where the request received in step S1101 is an unlock-with-PIN request, the unlock-with-PIN response includes information indicating that unlocking with the PIN code is successful.

In step S1160, the cooperation control unit 503 transmits to the data processing apparatus 101 as a response to the unlock-with-PIN request received in step S1101, an unlock-with-PIN response that includes information indicating that unlocking fails because the PIN code does not match, and ends the process in the flowchart.

In step S1170, the cooperation control unit 503 transmits to the data processing apparatus 101 as a response to the cooperation state obtaining request received in step S1101, a cooperation state obtaining response that includes information about the current cooperation state, and ends the process in the flowchart.

In step S1180, the cooperation control unit 503 transmits to the data processing apparatus 101 as a response to the request received in step S1101, an error response that includes information indicating that the request is not accepted, and ends the process in the flowchart.

With the process described above, the image forming apparatus 104 can reset the automatic clearing timer when accepting a request from the data processing apparatus 101 without the operation unit 307 accepting an operation. Therefore, for example, the PIN code screen (FIG. 12E) is not prevented from being displayed, and the user can make the process proceed. The automatic clearing timer is reset only in a case where an accepted request from the data processing apparatus 101 is a request based on an operation

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on the data processing apparatus 101, and therefore, the automatic clearing timer is not reset undesirably. For example, a cooperation state obtaining request is a request that is transmitted occasionally without the user operating the data processing apparatus 101, and therefore, when receiving this request, the image forming apparatus 104 does not reset the automatic clearing timer. The automatic clearing timer of the image forming apparatus 104 may be configured to reset each time one character of a PIN code is input on the data processing apparatus 101.

With reference to FIG. 13, a process performed in a case where a user operation is performed on the image forming apparatus 104 will be described. FIG. 13 is a flowchart for explaining the process in the image forming apparatus 104, and the process is implemented by the CPU 302 of the image forming apparatus 104 loading a program for implementing a control module stored in the ROM 304 or the HDD 305 to the RAM 303 and executing the program. The process in the flowchart illustrated in FIG. 13 is performed asynchronously to the process in the flowchart illustrated in FIGS. 11A and 11B.

In step S1301, the cooperation control unit 503 detects an event occurring in the image forming apparatus 104. For example, when the user places an original document on the scanner 313, a “scanner: an original document placed” event occurs. When the user removes an original document from the scanner 313, a “scanner: no original document placed” event occurs. When the user presses the placing completed button 1233 in FIG. 12D, a “UI: the placing completed button pressed” event occurs. When the user presses the cancel button 1232 or the cancel button 1241, a “UI: the cancel button pressed” event occurs.

In step S1302, the cooperation control unit 503 obtains from the event management table 650 information (information defined in the “occurring event” column 652 and in the “changed cooperation state” column 653) corresponding to the current cooperation state.

In step S1303, the cooperation control unit 503 determines whether the event detected in step S1301 is an event that is to be handled in the current cooperation state. Specifically, the cooperation control unit 503 determines whether the event detected in step S1301 is included in the “occurring event” column 652 obtained in step S1302. If the event is an event to be handled, the cooperation control unit 503 makes the process proceed to step S1304; otherwise the cooperation control unit 503 ends the process in the flowchart.

In step S1304, the cooperation control unit 503 resets the automatic clearing timer of the image forming apparatus 104.

In step S1305, the cooperation control unit 503 makes the process diverge in accordance with the “changed cooperation state” column 653 obtained in step S1302. If the changed cooperation state is “waiting for an original document to be placed”, the cooperation control unit 503 makes the process proceed to step S1306. If the changed cooperation state is “PIN code request”, the cooperation control unit 503 makes the process proceed to step S1310. If the changed cooperation state is “cancel”, the cooperation control unit 503 makes the process proceed to step S1320.

In step S1306, the cooperation control unit 503 changes the cooperation state to the “waiting for an original document to be placed” state.

In step S1307, the UI control unit 502 displays the original-document placing screen on the operation unit 307 of the image forming apparatus 104 and ends the process in the flowchart.

In step S1310, the cooperation control unit 503 changes the cooperation state to the "PIN code request" state. In step S1311, the cooperation control unit 503 generates a PIN code. In step S1312, the UI control unit 502 displays the PIN code screen on the operation unit 307 of the image forming apparatus 104 and ends the process in the flowchart.

In step S1320, the cooperation control unit 503 changes the cooperation state to the "cancel" state. In step S1321, the cooperation control unit 503 sets a timer for canceling the cooperation state and ends the process in the flowchart. After the time set for the timer has elapsed, the cooperation control unit 503 deletes the cooperation state and the job settings saved in step S1108, and the image forming apparatus 104 transitions to the "not in cooperation" state.

With reference to FIG. 14, a process for automatic clearing control in the image forming apparatus 104 will be described. FIG. 14 is a flowchart for explaining the process in the image forming apparatus 104, and the process is implemented by the CPU 302 of the image forming apparatus 104 loading a program for implementing a control module stored in the ROM 304 or the HDD 305 to the RAM 303 and executing the program. The process in the flowchart illustrated in FIG. 14 is performed asynchronously to the processes in the flowcharts illustrated in FIGS. 11A and 11B and FIG. 13.

Note that in this embodiment, an example case where the initial screen of the image forming apparatus 104 is displayed in response to the occurrence of automatic clearing in the image forming apparatus 104 will be described. The initial screen can be set by the user, or the home screen (FIG. 12B) may be set or a specific application screen may be specified. The operation performed in response to the occurrence of automatic clearing is an example, and another example operation may be performed in which in a case where the image forming apparatus 104 has an authentication function, the user is made to log out of the image forming apparatus 104.

In step S1401, the UI control unit 502 detects the image forming apparatus 104 returning from a sleep state. The sleep state is a state in which only limited components are energized to reduce power consumption in a case where the image forming apparatus 104 is not used for a predetermined time. When the user operates the image forming apparatus 104, the image forming apparatus 104 returns from the sleep state and becomes ready for use. Although the time during which the image forming apparatus 104 is in the sleep state may be longer than the automatic clearing time or equal to the automatic clearing time, in this embodiment, a description will be given under the assumption that the time during which the image forming apparatus 104 is in the sleep state is longer than the automatic clearing time.

In step S1402, in response to the return from the sleep state, the UI control unit 502 displays the initial screen on the operation unit 307 of the image forming apparatus 104. In step S1403, the UI control unit 502 starts counting down by the automatic clearing timer. Thereafter, the automatic clearing timer decreases the remaining time as time passes.

In step S1404, the UI control unit 502 monitors the remaining time of the automatic clearing timer, and makes the process proceed to step S1405 when detecting the remaining time being equal to zero or keeps monitoring the remaining time if the remaining time is more than zero.

In step S1405, the cooperation control unit 503 determines whether the image forming apparatus 104 is in cooperation with the data processing apparatus 101 currently. If the image forming apparatus 104 is in cooperation with the data processing apparatus 101, the cooperation

control unit 503 makes the process proceed to step S1406; otherwise the cooperation control unit 503 makes the process proceed to step S1407.

In step S1406, the cooperation control unit 503 deletes the cooperation state and the job settings saved in step S1108 and cancels cooperation with the data processing apparatus 101.

In step S1407, the UI control unit 502 displays the initial screen on the operation unit 307 of the image forming apparatus 104 and ends the process in the flowchart.

With the procedure described in this embodiment, in a case where a user operation is performed on either the data processing apparatus 101 or the image forming apparatus 104 while the data processing apparatus 101 and the image forming apparatus 104 are operating in cooperation with each other, the automatic clearing timers of both apparatuses are reset. Accordingly, the automatic clearing function appropriately works while the data processing apparatus 101 and the image forming apparatus 104 are operating in cooperation with each other, resulting in increased usability.

Second Embodiment

In the first embodiment, in a case where automatic clearing occurs in one of the data processing apparatus 101 or the image forming apparatus 104, automatic clearing is made to occur in the other apparatus in accordance with the automatic clearing time of the other apparatus. In a second embodiment, an example configuration in which in a case where automatic clearing occurs in one of the apparatuses, automatic clearing is made to occur also in the other apparatus will be described. The basic configuration of the second embodiment is the same as that of the first embodiment, and therefore, only differences will be described.

With reference to FIG. 15, a process for automatic clearing control performed by the data processing apparatus 101 in the second embodiment will be described. FIG. 15 is a flowchart obtained by modifying the flowchart in FIG. 10. Only differences from FIG. 10 will be described.

In step S1501, the MFP control unit 403 determines whether the data processing apparatus 101 is in cooperation with the image forming apparatus 104. Although determination may be performed in any manner, for example, the cooperation state obtaining response received in step S708 may be retained, and the cooperation state may be used to perform determination.

Alternatively, determination may be performed on the basis of whether the screen currently displayed on the display 214 is a screen that is displayed during cooperation. If the data processing apparatus 101 is in cooperation with the image forming apparatus 104, the MFP control unit 403 makes the process proceed to step S1502; otherwise the MFP control unit 403 makes the process proceed to step S1004.

In step S1502, the MFP control unit 403 transmits to the image forming apparatus 104 a notification of the occurrence of automatic clearing.

In step S1511, the MFP control unit 403 determines whether a notification of the occurrence of automatic clearing is received from the image forming apparatus 104. The MFP control unit 403 makes the process proceed to step S1512 if the notification is received, or makes the process return to step S1003 if the notification is not received.

In step S1512, the MFP control unit 403 determines whether the notification received in step S1511 is a notification from the image forming apparatus 104 that is in cooperation with the data processing apparatus 101. If the

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source of the notification is the image forming apparatus 104, the MFP control unit 403 makes the process proceed to step S1004; otherwise the MFP control unit 403 makes the process return to step S1003.

With reference to FIG. 16, a process for automatic clearing control performed by the image forming apparatus 104 in the second embodiment will be described. FIG. 16 is a flowchart obtained by modifying the flowchart in FIG. 14. Only differences from FIG. 14 will be described.

In step S1601, the cooperation control unit 503 transmits to the data processing apparatus 101 a notification of the occurrence of automatic clearing.

In step S1611, the cooperation control unit 503 determines whether a notification of the occurrence of automatic clearing is received from the data processing apparatus 101. The cooperation control unit 503 makes the process proceed to step S1612 if the notification is received, or makes the process return to step S1404 if the notification is not received.

In step S1612, the cooperation control unit 503 determines whether the notification received in step S1611 is a notification from the data processing apparatus 101 that is in cooperation with the image forming apparatus 104. If the source of the notification is the data processing apparatus 101, the cooperation control unit 503 makes the process proceed to step S1407; otherwise the cooperation control unit 503 makes the process return to step S1404.

As described above, according to the second embodiment, in a case where automatic clearing occurs in one of the apparatuses, a notification of the automatic clearing is transmitted to the other apparatus, and this can make automatic clearing occur in both apparatuses that are in cooperation, resulting in increased usability.

Other Embodiments

Embodiment(s) of the present invention can also be realized by a computer of a system or apparatus that reads out and executes computer executable instructions (e.g., one or more programs) recorded on a storage medium (which may also be referred to more fully as 'non-transitory computer-readable storage medium') to perform the functions of one or more of the above-described embodiment(s) and/or that includes one or more circuits (e.g., application specific integrated circuit (ASIC)) for performing the functions of one or more of the above-described embodiment(s), and by a method performed by the computer of the system or apparatus by, for example, reading out and executing the computer executable instructions from the storage medium to perform the functions of one or more of the above-described embodiment(s) and/or controlling the one or more circuits to perform the functions of one or more of the above-described embodiment(s). The computer may comprise one or more processors (e.g., central processing unit (CPU), micro processing unit (MPU)) and may include a network of separate computers or separate processors to read out and execute the computer executable instructions. The computer executable instructions may be provided to the computer, for example, from a network or the storage medium. The storage medium may include, for example, one or more of a hard disk, a random-access memory (RAM), a read only memory (ROM), a storage of distributed computing systems, an optical disk (such as a compact disc (CD), digital versatile disc (DVD), or Blu-ray Disc (BD)), a flash memory device, a memory card, and the like.

While the present invention has been described with reference to embodiments, it is to be understood that the

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invention is not limited to the disclosed embodiments, but is defined by the scope of the following claims.

What is claimed is:

1. An information processing apparatus configured to communicate with a terminal, the information processing apparatus comprising:

at least one processor and at least a memory coupled to the at least one processor and having instructions stored thereon, the at least one processor, by executing the instructions, acting as:

a control unit configured to execute automatic clearing for both the information processing apparatus and the terminal on a basis of a user operation on an operation unit of the information processing apparatus and a user operation on an operation unit of the terminal, wherein

the control unit executes the automatic clearing for both the information processing apparatus and the terminal in a case where there is no user operation for predetermined time on the operation unit of the information processing apparatus nor on the operation unit of the terminal, and the control unit does not execute the automatic clearing for the information processing apparatus in a case where there is a user operation on the operation unit of the terminal even if there is no user operation on the operation unit of the information processing apparatus before the predetermined time passes after there is no user operation on the operation unit of the information processing apparatus nor on the operation unit of the terminal.

2. The information processing apparatus according to claim 1, the at least one processor further acting as:

a counting unit configured to count time for which no operation is received from a user, wherein

the control unit executes the automatic clearing in a case where the time counted by the counting unit reaches the predetermined time,

the counting unit resets counting in a case where a user operation on the operation unit of the information processing apparatus is received, and

the counting unit resets counting in a case where a user operation on the operation unit of the terminal is received.

3. The information processing apparatus according to claim 1,

wherein the control unit executes the automatic clearing on a basis of an automatic clearing timer, and

wherein the automatic clearing timer is based on a user operation on the operation unit of the information processing apparatus and a user operation on the operation unit of the terminal.

4. The information processing apparatus according to claim 1, wherein the automatic clearing process includes a process of displaying an initial screen.

5. The information processing apparatus according to claim 1, wherein the automatic clearing process includes a process of clearing a value of a setting set on a displayed screen.

6. The information processing apparatus according to claim 1, wherein the automatic clearing process includes a process of making a logged-in user of the information processing apparatus log out of the information processing apparatus.

7. The information processing apparatus according to claim 1, wherein the information processing apparatus is an

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image forming apparatus having at least one of a print function, a scan function, or a copy function.

8. The information processing apparatus according to claim 1, wherein the user operation on the operation unit of the terminal is a user operation received by an application that cooperates with the information processing apparatus and runs on the terminal. 5

9. The information processing apparatus according to claim 1, wherein the user operation on the operation unit of the terminal is an operation of selecting a button displayed by an application that cooperates with the information processing apparatus and runs on the terminal. 10

10. The information processing apparatus according to claim 9, wherein the application is an application that gives a print instruction to the information processing apparatus. 15

11. A method for an information processing apparatus configured to communicate with a terminal, the method comprising:

executing automatic clearing for both the information processing apparatus and the terminal on a basis of a user operation on an operation unit of the information processing apparatus and a user operation on an operation unit of the terminal, wherein 20

the automatic clearing is executed for both the information processing apparatus and the terminal in a case where there is no user operation for predetermined time on the operation unit of the information processing apparatus nor on the operation unit of the terminal, and the automatic clearing is not executed for the information processing apparatus in a case where there is a user 25

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operation on the operation unit of the terminal even if there is no user operation on the operation unit of the information processing apparatus before the predetermined time passes after there is no user operation on the operation unit of the information processing apparatus nor on the operation unit of the terminal.

12. A non-transitory computer-readable storage medium storing executable instructions, which when executed by one or more processors of an information processing apparatus, cause the information processing apparatus to perform operations comprising:

executing automatic clearing for both the information processing apparatus and the terminal on a basis of a user operation on an operation unit of the information processing apparatus and a user operation on an operation unit of the terminal, wherein

the automatic clearing is executed for both the information processing apparatus and the terminal in a case where there is no user operation for predetermined time on the operation unit of the information processing apparatus nor on the operation unit of the terminal, and the automatic clearing is not executed for the information processing apparatus in a case where there is a user operation on the operation unit of the terminal even if there is no user operation on the operation unit of the information processing apparatus before the predetermined time passes after there is no user operation on the operation unit of the information processing apparatus nor on the operation unit of the terminal.

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